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Zero shot health trajectory prediction using transformer

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Pawel Renc [1,2,3], Yugang Jia [4], Anthony E. Samir [1,2], Jaroslaw Was[3], Quanzheng Li[1,2], David W. Bates [2,5,6] & Arkadiusz Sitek [1,2] ✉

Integrating modern machine learning and clinical decision-making has great promise for mitigating healthcare's increasing cost and complexity. We introduce the Enhanced Transformer for Health Outcome Simulation (ETHOS), a novel application of the transformer deep-learning architecture for analyzing high-dimensional, heterogeneous, and episodic health data. ETHOS is trained using Patient Health Timelines (PHTs)—detailed, tokenized records of health events—to predict future health trajectories, leveraging a zero-shot learning approach. ETHOS represents a significant advancement in foundation model development for healthcare analytics, eliminating the need for labeled data and model fine-tuning. Its ability to simulate various treatment pathways and consider patient-specific factors positions ETHOS as a tool for care optimization and addressing biases in healthcare delivery. Future developments will expand ETHOS' capabilities to incorporate a wider range of data types and data sources. Our work demonstrates a pathway toward accelerated AI development and deployment in healthcare.

Healthcare in the U.S. is the world's most expensive, and the quality and safety of care do not compare well to other developed countries[1]. While electronic healthcare records are now ubiquitous in the U.S., and decision-support technologies are widely implemented, most are rule-based, and their effectiveness so far has been limited[2]. Artificial intelligence has emerged as a technique with great potential for improving care, but most organizations are not using it to any major degree. Two major limiting factors have been (1) the lack of large, labeled datasets, which are expensive and time-consuming to develop; and (2) limited system capacity to deliver recommendations to the appropriate clinician at the optimal time. In this manuscript, we describe a novel method called the Enhanced Transformer for Health Outcome Simulation (ETHOS), which we believe can help address many of the limitations that have prevented widespread AI adoption.

ETHOS is a novel application of the transformer deep-learning architecture, originally conceptualized for natural language processing[3]. This architecture, a cornerstone in large language model (LLM) development, is repurposed in ETHOS to analyze health-related data, moving beyond the textual focus of traditional LLMs. ETHOS is designed to process Patient Health Timelines (PHTs)—detailed tokenized chronological records of health-related events—to predict future health timelines. In PHTs, a token serves as the fundamental unit of information, encapsulating diverse data types such as patient admissions, administered medications, or time intervals. We elaborate on this pivotal aspect of our methodology in the Methods section. Our model takes the patient's health history, as represented by PHT, and subsequently forecasts future PHT (fPHT) on a token-by-token basis (refer to Fig. 1).

ETHOS's generative capabilities are gained in unsupervised learning. Once trained, ETHOS can forecast future health events without requiring task-specific training. This is done through a zero-shot learning approach, making ETHOS a versatile foundation model for numerous healthcare applications. With appropriate modifications, ETHOS can be adapted to a broad range of data types, including but not limited to medical images, clinical and discharge notes, monitoring data, data from wearables, or omics data.

In this research, we leverage the recently released MIMIC-IV v.2.2 dataset[4,5], a rich open-source repository accompanied by our code, allowing others to replicate our findings. MIMIC-IV is expansive, chronicling >400,000 hospitalizations in >200,000 patients. Although relatively large, we anticipate that the performance of our system will further improve as we expand the dataset with additional patient histories and data types.

Importantly, we utilize the MIMIC-IV dataset in its original noisy form without any data modifications, cleaning, or targeted imputation for missing entries. The information is retained in the face of large data inconsistencies, such as discharge dates noted before admission dates. We operated under the assumption that, within large enough datasets and appropriate

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使用变压器进行零射击健康轨迹预测

检查更新

Pawel Renc [1,2]、Yugang Jia [4]、Anthony E. Samir [1,2]、Jaroslaw Was[3]、Quanzheng Li[1,2]、David W. Bates [2,5,6] 和 Arkadiusz Sitek [1,2]

将现代机器学习和临床决策相结合对于缓解医疗保健日益增加的成本和复杂性具有巨大的希望。我们介绍了用于健康结果模拟的增强型 Transformer (ETHOS)，这是 Transformer 深度学习架构的一种新颖应用，用于分析高维、异构和情景健康数据。 ETHOS 使用患者健康时间线 (PHT)（详细的、标记化的健康事件记录）进行训练，利用零样本学习方法来预测未来的健康轨迹。 ETHOS 代表了医疗保健分析基础模型开发的重大进步，消除了对标记数据和模型微调的需求。 ETHOS 能够模拟各种治疗途径并考虑患者特定因素，使其成为优化护理和解决医疗保健服务偏差的工具。未来的发展将扩展 ETHOS 的能力，以纳入更广泛的数据类型和数据源。我们的工作展示了一条加速医疗保健领域人工智能开发和部署的途径。

美国的医疗保健是世界上最昂贵的，而且医疗质量和安全性与其他发达国家相比并不好[1]。虽然电子医疗记录现在在美国无处不在，决策支持技术也得到了广泛应用，但大多数都是基于规则的，迄今为止其有效性有限[2]。人工智能已经成为一种在改善护理方面具有巨大潜力的技术，但大多数组织并未在很大程度上使用它。两个主要限制因素是（1）缺乏大型标记数据集，这些数据集的开发成本昂贵且耗时； (2) 系统在最佳时间向适当的临床医生提供建议的能力有限。在这篇手稿中，我们描述了一种称为健康结果模拟增强型变压器（ETHOS）的新方法，我们相信它可以帮助解决阻碍人工智能广泛采用的许多限制。

ETHOS 是 Transformer 深度学习架构的一种新颖应用，最初是为自然语言处理而概念化的[3]。该架构是大语言模型 (LLM) 开发的基石，在 ETHOS 中被重新用于分析健康相关数据，超越了传统 LLM 的文本焦点。 ETHOS 旨在处理患者健康时间表 (PHT)（健康相关事件的详细标记化时间顺序记录），以预测未来的健康时间表。在 PHT 中，令牌作为信息的基本单元，封装了不同的数据类型，例如患者入院、服用的药物或时间间隔。我们在“方法”部分详细阐述了我们方法论的这一关键方面。我们的模型采用患者的健康史（用 PHT 表示），然后逐个预测未来的 PHT (fPHT)（参见图 1）。

ETHOS 的生成能力是在无监督学习中获得的。经过培训后，ETHOS 可以预测未来的健康事件，而无需进行特定任务的培训。这是通过零样本学习方法完成的，使 ETHOS 成为众多医疗保健应用程序的多功能基础模型。通过适当的修改，ETHOS 可以适应广泛的数据类型，包括但不限于医学图像、临床和出院记录、监测数据、可穿戴设备数据或组学数据。

在这项研究中，我们利用最近发布的 MIMIC-IV v.2.2 数据集[4,5]，这是一个丰富的开源存储库以及我们的代码，允许其他人复制我们的研究结果。 MIMIC-IV 内容广泛，记录了 >200,000 名患者的 400,000 次住院治疗。尽管相对较大，但我们预计，随着我们使用更多患者历史和数据类型扩展数据集，系统的性能将进一步提高。

重要的是，我们以其原始的噪声形式使用 MIMIC-IV 数据集，没有任何数据修改、清理或对缺失条目的有针对性的插补。面对大量数据不一致的情况，例如在入院日期之前记录的出院日期，该信息仍被保留。我们的操作假设是，在足够大的数据集和适当的情况下

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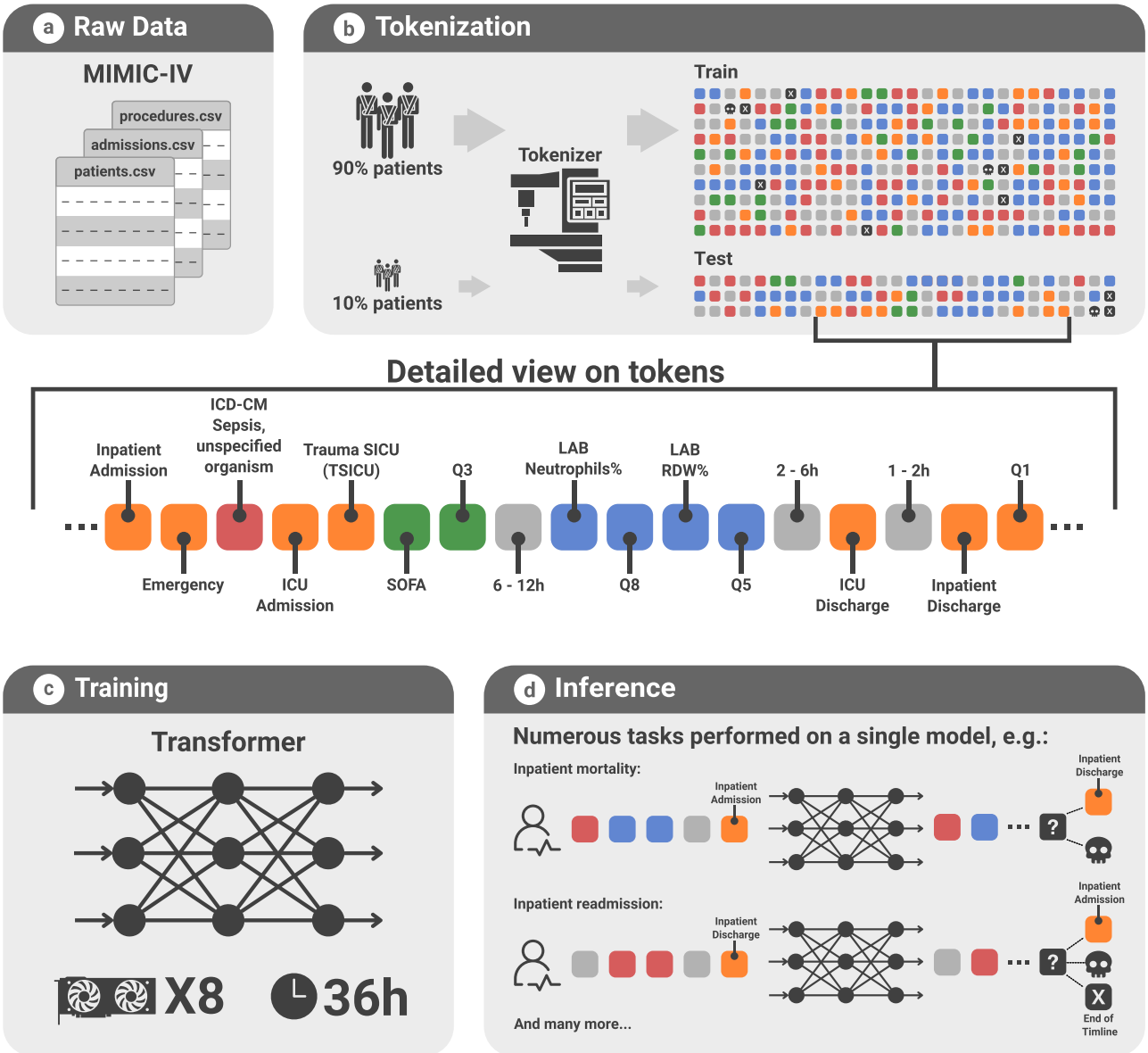

Fig. 1 | Implementing the ETHOS model with EMR data. a Extraction of raw patient data from the MIMIC-IV database, encompassing tables of admissions, patient demographics, medical procedures, among others. **b** The tokenization process, utilizing data from 90% of patients for model training and the remaining 10% for testing, transforms complex medical records into structured PHT for efficient model processing. **c** Training phase illustration, employing a transformer architecture optimized across 8 GPUs over a span of 36 h. **d** Demonstration of ETHOS's zero-shot inference capabilities, highlighting its proficiency in performing tasks such as predicting inpatient mortality and readmission rates, leveraging forecasted future PHTs.

tokenization and training methods, ETHOS would be robust enough to handle the noisy input and automatically manage the noise/anomalies in the input data. The resilience of ETHOS to data inaccuracies and missing information has important implications for the efficiency of downstream model development. Healthcare data inevitably contains errors, some of which may not be immediately apparent or easily rectifiable. Attempts to clean large datasets can be impractical and may inadvertently introduce biases and errors. Our approach highlights the vital need for algorithms adept at managing these challenges, a prerequisite for the large-scale development of reliable and robust healthcare AI applications.

Our research showcases the zero-shot learning capabilities of ETHOS in predicting inpatient and ICU mortality, estimating ICU length of stay (LOS), and determining readmission probabilities. Additionally, we illustrate the model's versatility by performing a regression task to estimate the first-day Sequential Organ Failure Assessment (SOFA) score[4,6] at the time of ICU admission using information before admission (see example in Fig. 1d). The SOFA score is a critical tool for monitoring a patient's condition in the ICU, evaluating organ function or failure across six systems—respiratory, cardiovascular, hepatic, coagulation, renal, and neurological—with each system scored from 0 to 4, culminating in a total possible minimum score of 0 and maximum score of 24. Furthermore, we predict Diagnostic-Related Group (DRG) classifications, encompassing over 771 categories, at the time of hospital discharge. The DRG system categorizes hospital cases into standardized case complexity-based Medicare and Medicaid payment groups, encouraging efficient patient care without compromising quality. The diversity of tasks ETHOS can perform, from mortality predictions and LOS estimation to SOFA scoring and DRG classification, highlights its broad applicability and zero-shot learning efficiency.

ETHOS is a foundation model[7], introducing a novel approach in the landscape of data analysis within the healthcare domain. The other foundational models developed recently have fallen into two broad categories. The first of these categories encompasses Clinical Language Models

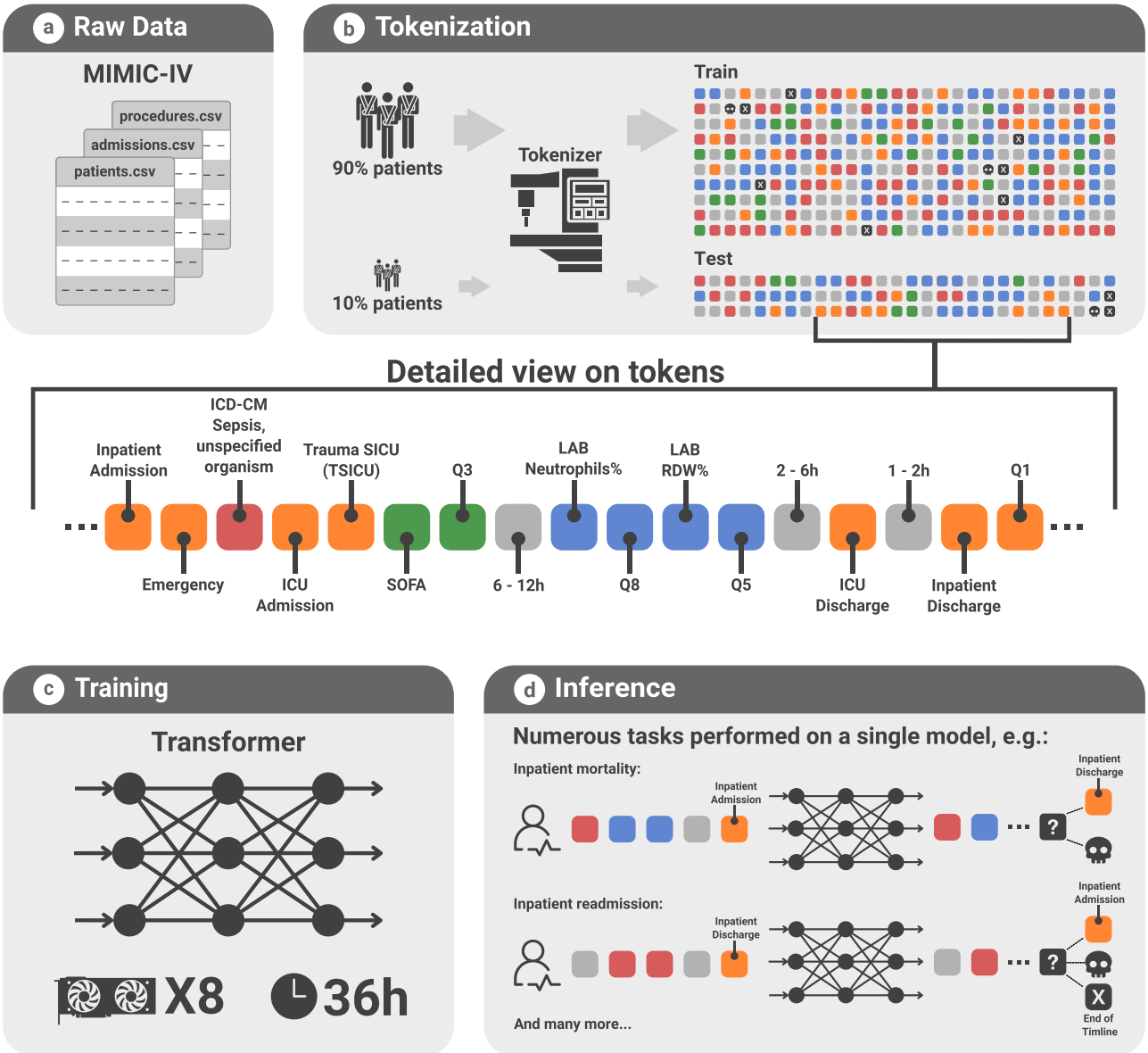

图 1 | 使用 EMR 数据实现 ETHOS 模型。 a 从 MIMIC-IV 数据库中提取原始患者数据，包括入院表、患者人口统计数据、医疗程序等。 b 标记化过程利用 90% 患者的数据进行模型训练，其余 10% 进行测试，将复杂的医疗记录转换为结构化 PHT，以实现高效的模型处理。 c 训练阶段图示，采用在 36 小时内跨 8 个 GPU 进行优化的 Transformer 架构。 d 展示 ETHOS 的零样本推理能力，强调其在执行预测住院患者死亡率和再入院率等任务方面的熟练程度，利用预测的未来 PHT。

通过标记化和训练方法，ETHOS 将足够强大，能够处理噪声输入并自动管理输入数据中的噪声/异常。 ETHOS 对数据不准确和信息缺失的恢复能力对于下游模型开发的效率具有重要影响。医疗保健数据不可避免地包含错误，其中一些错误可能不会立即显现或容易纠正。尝试清理大型数据集可能不切实际，并且可能会无意中引入偏差和错误。我们的方法强调了对善于应对这些挑战的算法的迫切需求，这是大规模开发可靠且强大的医疗保健人工智能应用程序的先决条件。

我们的研究展示了 ETHOS 在预测住院患者和 ICU 死亡率、估计 ICU 住院时间 (LOS) 以及确定再入院概率方面的零样本学习能力。此外，我们通过执行回归任务来使用入院前的信息估计入ICU时的第一天序贯器官衰竭评估（SOFA）评分[4,6]来说明该模型的多功能性（参见图1d中的示例）。

SOFA 评分是监测 ICU 患者病情的重要工具，评估六个系统（呼吸、心血管、肝脏、凝血、肾脏和神经系统）的器官功能或衰竭，每个系统的评分从 0 到 4 分，最终可能的最低总分为 0 分，最高分为 24 分。此外，我们预测出院时的诊断相关组 (DRG) 分类，涵盖超过 771 个类别。 DRG 系统将医院病例分类为基于标准化病例复杂性的医疗保险和医疗补助支付组，鼓励在不影响质量的情况下进行高效的患者护理。 ETHOS 可以执行的任务多样性，从死亡率预测和 LOS 估计到 SOFA 评分和 DRG 分类，凸显了其广泛的适用性和零样本学习效率。

ETHOS 是一个基础模型[7]，在医疗保健领域的数据分析领域引入了一种新颖的方法。最近开发的其他基础模型分为两大类。第一个类别包括临床语言模型

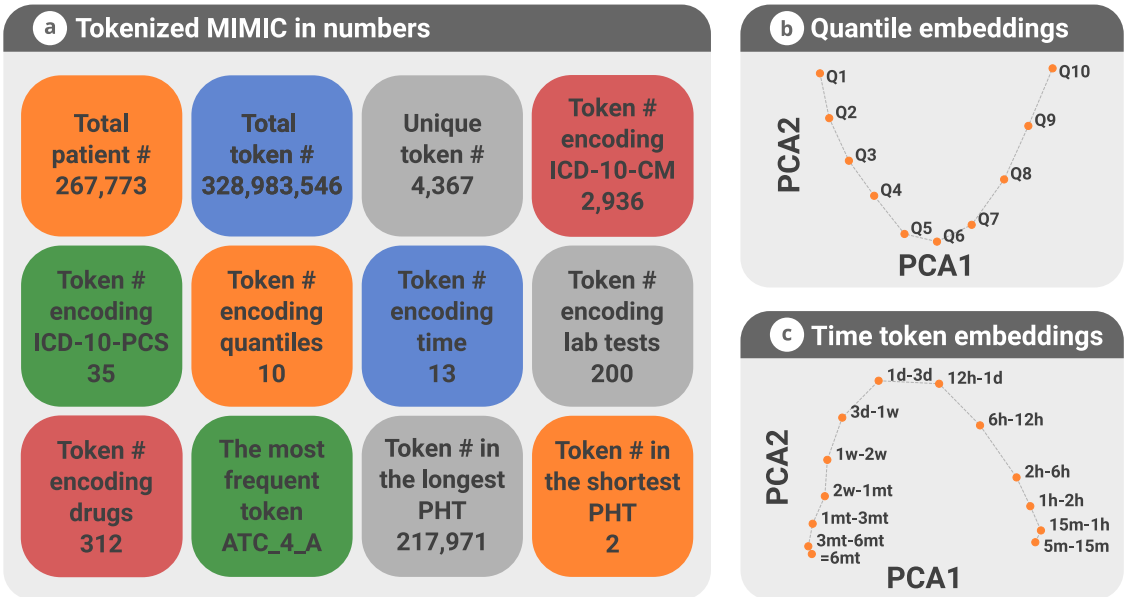

Fig. 2 | Tokenization and embedding visualizations of MIMIC-IV data. **a** Overview of key insights derived from the tokenization process applied to MIMIC-IV data. **b** Visualization of embedding vectors for quantile tokens (Qs), which categorize quantitative information across the dataset. Each quantitative measure (e.g., blood pressure) is encoded by a preceding category-specific token followed by a quantile token, delineating its position within a predefined value range. This method facilitates a structured, scalable representation of complex data types via a systematic token sequence. **c** Visualization of embedding vectors for time-interval tokens, illustrating the temporal distribution and relationships within the PHT.

(CLaMs), a specialized subset of large language models (LLMs)[8] tailored for processing clinical and biomedical text data. These models are typically trained on extensive datasets containing clinical notes, biomedical literature, and other healthcare-related text sources. CLaMs are proficient in various clinical tasks such as extracting drug names, summarizing medical dialogues, predicting clinical outcomes, and responding to patient queries[9–13]. The second category comprises Foundation Models for Electronic Medical Records (FEMRs), representing another class of clinical foundation models tailored specifically for EMR data analysis. FEMRs undergo training on the extensive medical histories of patients, covering both structured data (such as demographics and lab results) and unstructured data (including progress notes and radiology reports). Unlike CLaMs, FEMRs are not designed to generate clinical text. Instead, they produce machine-understandable representations of patient data, facilitating tasks such as patient phenotyping and outcome prediction[12,14–16]. Similarly, data that chronicles human lives, akin to EMR, can also be modeled effectively in this manner[12,14–16].

The primary distinction between ETHOS and previously published methods lies in our approach, which eliminates the need for fine-tuning or labeled data to produce accurate inferences or predictions. We demonstrate inference across a wide array of tasks without task-specific training. Moreover, the ability of ETHOS to forecast future PHTs opens the door to a wide array of bespoke and innovative applications, facilitating its use in unique scenarios in healthcare, some of them explored in the discussion section. Unlike many studies, which often apply specific criteria for selecting data for training and testing, our methodology imposes no such limitations. This feature is crucial for considering the scalability of the ETHOS approach to data sets comprising millions or even hundreds of millions of patients.

Results

Tokenization of MIMIC data and training of ETHOS

Figure 2a summarizes some statistics of the tokenization process, including the number of tokens generated and other details. Supplementary Fig. 3 presents visualizations of the 768-dimensional embeddings reduced to a 2D plane using Principal Component Analysis (PCA) for quantile tokens, which encode all quantitative values in the data. The tokens are arranged from Q1 (the lowest quantile) to Q10 (the highest quantile). This suggests that the transformer model has learned a sequential relationship between the tokens that mirrors their natural order, ascertaining this order from the data during the training process. The proximity between points could reflect the model's differentiation among the quantiles. We observe that the gaps between Q4, Q5, and Q6 are narrower than those between Q9 and Q10. This may suggest that the model deems the variance between population-average values to be less substantial than that of extremely high values. For example, the difference in clinical significance between a blood pressure reading of 110 mmHg (Q5) and one of 130 mmHg (Q6) is less pronounced than the difference between 140 mmHg (Q9) and 160 mmHg (Q10), which could account for the greater disparity in the embedding vectors of high quantiles.

The embeddings for time-interval tokens, representing the approximate durations between different tokenized events in PHT, are illustrated in Supplementary Fig. 3b. These embeddings display a pattern analogous to that observed for Q tokens, where ETHOS systematically arranged them according to the actual time values they represent. Remarkably, the model perceives the two shortest (5m-15m, 15m-1h), and two longest (3m-6m, 6 m) intervals as relatively similar.

ETHOS inferences

In our study, we conducted zero-shot inferences for a diverse array of classification tasks, including readmission to the ICU, inpatient mortality, ICU mortality, combined inpatient and ICU mortality in patients with sepsis, readmission to the ICU for patients with intracerebral hemorrhage, assignment of DRG class assessed at inpatient discharge. We also demonstrate regression of first-day SOFA score at the time of ICU admission and regression of the length of stay in ICU in days assessed upon admission. The results corresponding to these tasks are summarized in Fig. 3. We also provide precision-recall curves of the corresponding results in Supplementary Fig. 7.

To situate our results within the broader scientific discourse, we conducted a literature review, concentrating on contemporary studies that utilized the MIMIC-III and MIMIC-IV datasets for similar tasks and reported their outcomes. A notable observation from our review is that many of these studies either lacked publicly available source code or implemented specific exclusion criteria for their data selection. Such practices pose challenges for directly comparing their results with our approach. Nonetheless, we posit that the numerical outcomes reported in these works

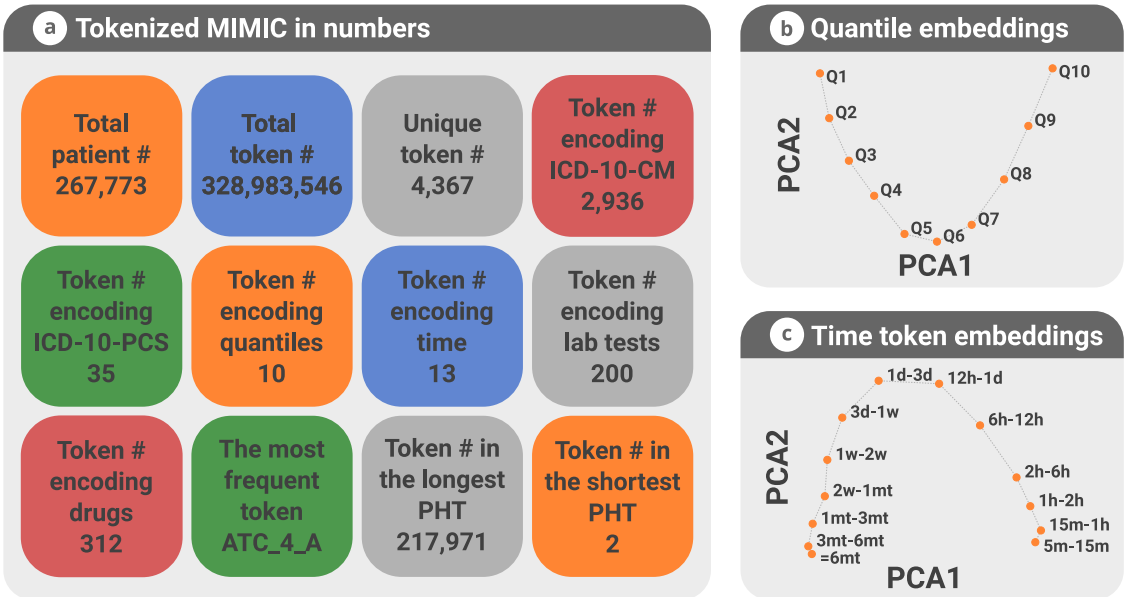

图 2 | MIMIC-IV 数据的标记化和嵌入可视化。 a 概述从应用于 MIMIC-IV 数据的标记化过程得出的关键见解。 b 分位数标记 (Qs) 嵌入向量的可视化，对整个数据集中的定量信息进行分类。每个定量测量（例如血压）都由前面的类别特定标记进行编码，后跟分位数标记，在预定义的值范围内描绘其位置。该方法通过系统的标记序列促进复杂数据类型的结构化、可扩展的表示。 c 时间间隔标记嵌入向量的可视化，说明 PHT 内的时间分布和关系。

(CLaMs)，大型语言模型 (LLM)[8] 的专门子集，专为处理临床和生物医学文本数据而定制。这些模型通常在包含临床记录、生物医学文献和其他医疗保健相关文本源的广泛数据集上进行训练。 CLaM 精通各种临床任务，例如提取药物名称、总结医学对话、预测临床结果以及回答患者询问[9–13]。第二类包括电子病历基础模型 (FEMR)，代表另一类专门为 EMR 数据分析定制的临床基础模型。 FEMR 接受有关患者广泛病史的培训，涵盖结构化数据（例如人口统计和实验室结果）和非结构化数据（包括进度说明和放射学报告）。与 CLaM 不同，FEMR 并非旨在生成临床文本。相反，它们生成机器可理解的患者数据表示，促进患者表型分析和结果预测等任务[12,14–16]。同样，记录人类生活的数据（类似于 EMR）也可以通过这种方式进行有效建模[12,14–16]。

ETHOS 和之前发布的方法之间的主要区别在于我们的方法，它不需要微调或标记数据来产生准确的推论或预测。我们无需特定任务的训练即可在各种任务中展示推理能力。此外，ETHOS 预测未来 PHT 的能力为各种定制和创新应用打开了大门，促进其在医疗保健中独特场景中的使用，其中一些场景已在讨论部分中探讨。与许多通常采用特定标准来选择训练和测试数据的研究不同，我们的方法没有施加这样的限制。这一功能对于考虑 ETHOS 方法对包含数百万甚至数亿患者的数据集的可扩展性至关重要。

结果

MIMIC数据代币化和ETHOS训练

图 2a 总结了代币化过程的一些统计数据，包括生成的代币数量和其他详细信息。补充图 3 展示了使用分位数标记的主成分分析 (PCA) 简化为 2D 平面的 768 维嵌入的可视化，该分位数标记对数据中的所有定量值进行编码。标记从 Q1（最低分位数）到 Q10（最高分位数）排列。这表明 Transformer 模型已经学会了之间的顺序关系反映其自然顺序的标记，在训练过程中从数据中确定该顺序。点之间的接近程度可以反映模型在分位数之间的差异。我们观察到 Q4、Q5 和 Q6 之间的差距比 Q9 和 Q10 之间的差距更窄。这可能表明该模型认为总体平均值之间的方差没有极高值之间的方差大。例如，110 mmHg (Q5) 和 130 mmHg (Q6) 的血压读数之间的临床意义差异不如 140 mmHg (Q9) 和 160 mmHg (Q10) 之间的差异明显，这可以解释高分位数的嵌入向量中更大的差异。

时间间隔标记的嵌入表示 PHT 中不同标记化事件之间的近似持续时间，如补充图 3b 所示。这些嵌入显示的模式类似于 Q 代币观察到的模式，其中 ETHOS 根据它们所代表的实际时间值系统地排列它们。值得注意的是，该模型认为两个最短间隔（5m-15m、15m-1h）和两个最长间隔（3m-6m、6 m）相对相似。

精神推论

在我们的研究中，我们对各种分类任务进行了零样本推断，包括再入 ICU、住院患者死亡率、ICU 死亡率、败血症患者的住院患者和 ICU 合并死亡率、脑出血患者再入 ICU、住院患者出院时评估的 DRG 类别分配。我们还展示了入住 ICU 时第一天 SOFA 评分的回归以及入院时评估的 ICU 住院天数的回归。与这些任务相对应的结果总结在图 3 中。我们还在补充图 7 中提供了相应结果的精确召回曲线。

为了将我们的结果置于更广泛的科学论述中，我们进行了文献综述，重点关注利用 MIMIC-III 和 MIMIC-IV 数据集执行类似任务的当代研究，并报告了其结果。我们的审查中一个值得注意的观察结果是，许多研究要么缺乏公开的源代码，要么对其数据选择实施了特定的排除标准。这种做法对直接将其结果与我们的方法进行比较提出了挑战。尽管如此，我们认为这些作品中报告的数字结果

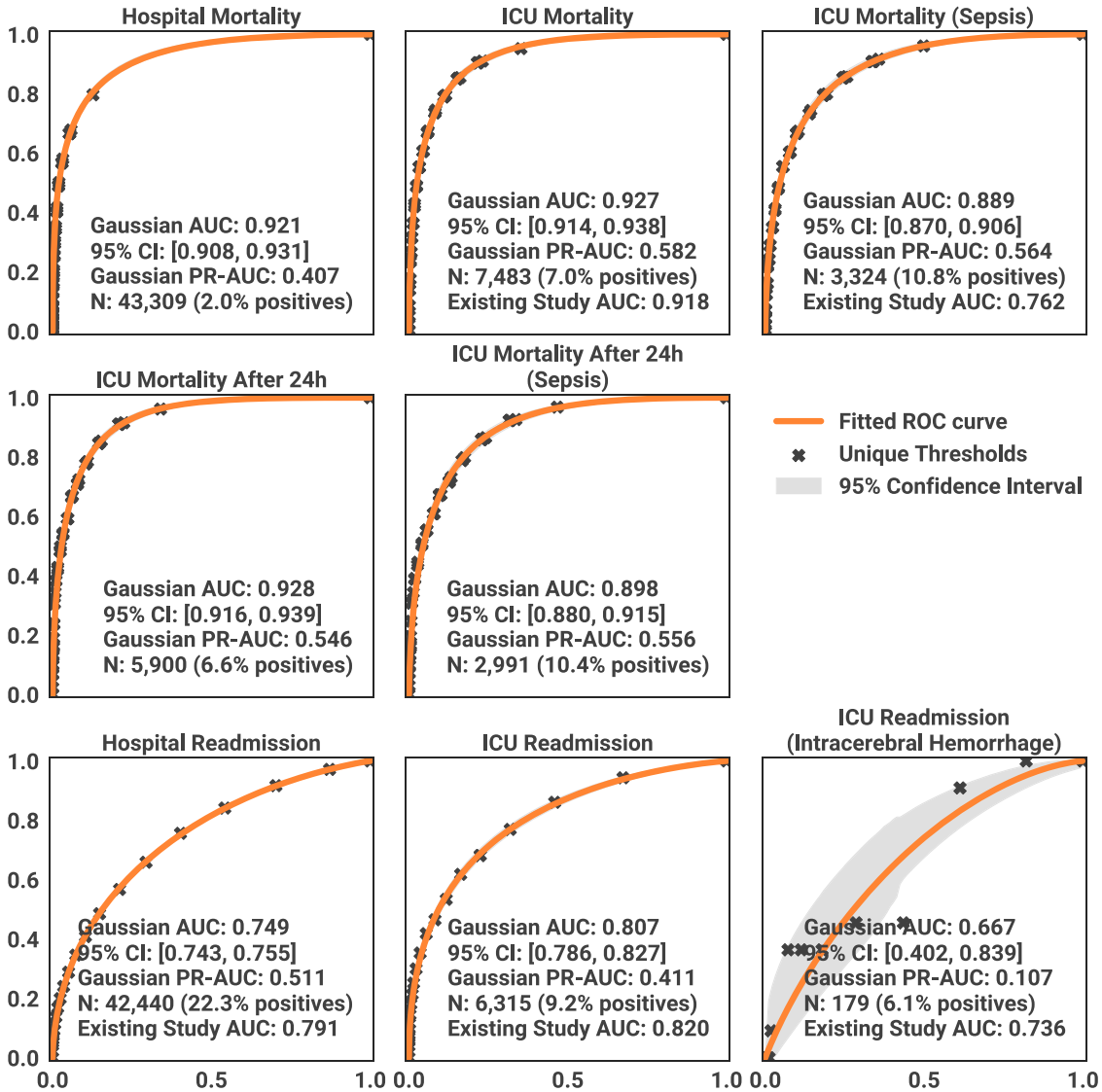

Fig. 3 | Receiver Operating Characteristic (ROC) curves for predictive tasks via the ETHOS model. Each graph delineates the model's efficacy in forecasting distinct clinical outcomes, specifically mortality and readmission rates. Accompanying each ROC curve are the case count (N), the outcome prevalence, and the 95% confidence interval for the AUC. Points marked with an 'X' denote specific thresholds utilized for classification decisions within the ETHOS model. Area under precision-recall (PR) curves is also provided and PR-curves are presented in supplementary material. The AUC of the existing study represents the performance of the best algorithms identified in the literature, with references provided within the text.

provide a valuable benchmark for assessing the performance of ETHOS. Furthermore, we conducted a direct comparative analysis of ETHOS against specialized algorithms developed in-house, with these findings detailed in the supplementary materials.

We conducted an analysis focusing on risk estimation for inpatient and ICU mortality, calculated at the respective points of patient admission to the hospital and ICU. The test set comprised 43,309 hospital admissions with a 2.0% mortality and 7483 ICU admissions with a 7.0% mortality. The ETHOS model demonstrated robust performance, achieving an AUC of 0.921 (95% CI: 0.908-0.931) for hospital mortality and 0.927 (95% CI: 0.914-0.938) for ICU mortality. Comparatively, in the ICU mortality risk prediction domain, the highest performance identified in our literature review was an AUC of 0.918 (95% CI: 0.915-0.922) reported by Pang et al.[17] using the XGBoost model. On the lower end, Chen et al.[18] reported an AUC of 0.642 ± 0.101. Within a specific subgroup of the test set of 3,324 patients with sepsis with 10.8% mortality prevalence, ETHOS's prediction of ICU mortality exhibited an AUC of 0.889 (95% CI: 0.870-0.906), which is a better performance than obtained in a study by Pan et al.[19], which estimated ICU mortality in adult sepsis patients using SOFA and additional features, achieving an AUC of 0.762 ± 0.006. We also estimated performance for a task of ICU mortality estimation for patients remaining in ICU for at least 24 h in which we obtained an AUC of 0.928 (95% CI: 0.916-0.939).

Furthermore, ETHOS estimated the length of stay (LoS) in the ICU with a mean absolute error (MAE) of 2.262 days (95% CI: 2.161–2.355 days). These results paralleled those of[18], who reported an MAE of 2.42 ± 0.10 days. ICU LoS prediction and mortality risk, underscoring the competitive zero-shot performance of ETHOS across multiple key healthcare metrics.

For the ICU readmission task, ETHOS' AUC of 0.807 (95% CI: 0.786-0.827) is slightly smaller than the AUC of 0.82 obtained using knowledge graph embeddings[20] and is higher than the AUC of 0.791 (95% CI, 0.782–0.800) using LSTMs based on MIMIC-III data[21]. Additionally, we applied our method to a task characterized by a relatively low prevalence, specifically focusing on only 174 cases of patients with hemorrhage admitted to the ICU present within our test set. The prediction of readmission by ETHOS yielded an AUC of 0.667 (95% CI: 0.402-0.839), comparable to the

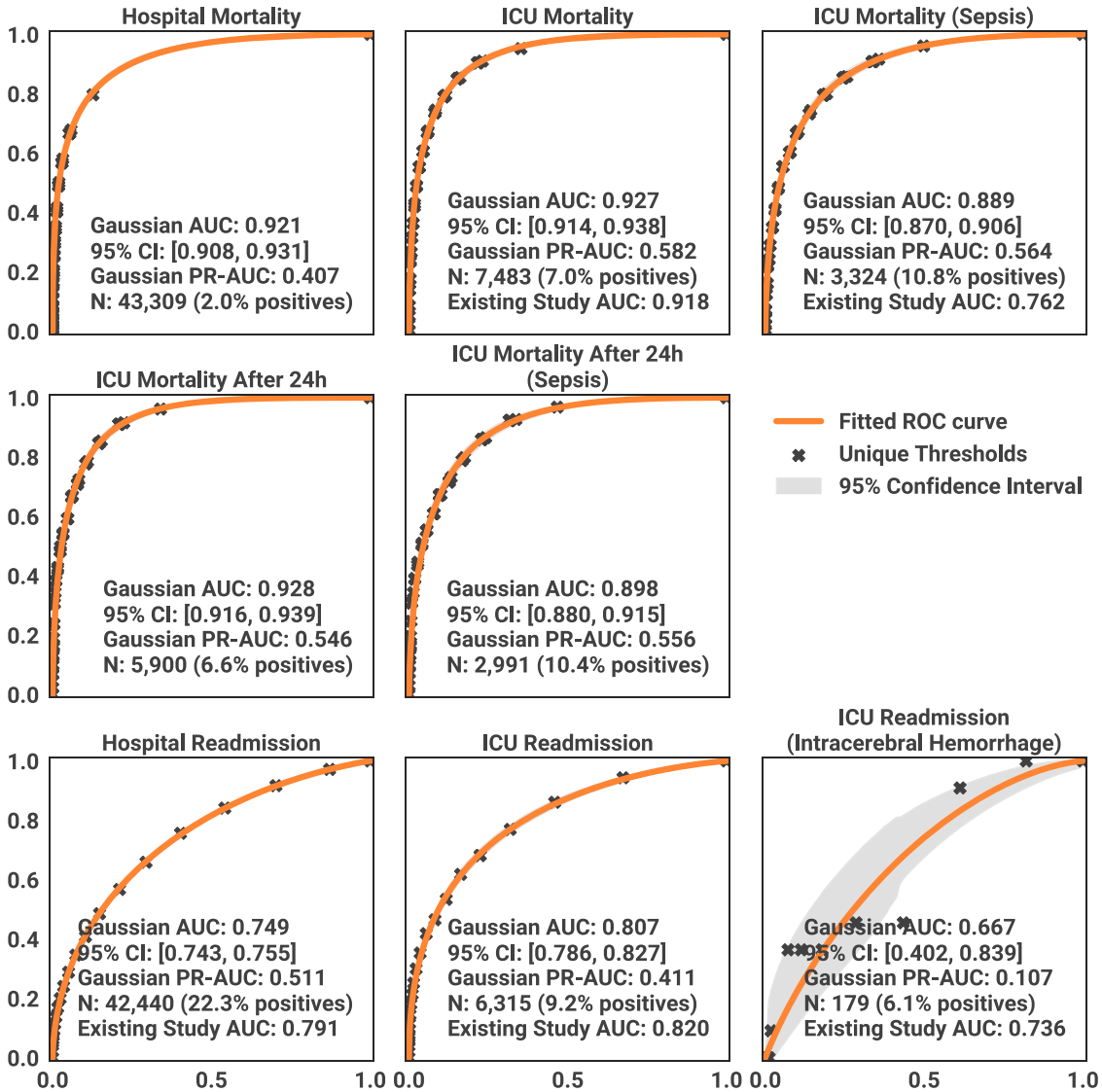

图 3 | 通过 ETHOS 模型进行预测任务的受试者工作特征 (ROC) 曲线。每张图都描绘了该模型在预测不同临床结果（特别是死亡率和再入院率）方面的功效。每条 ROC 曲线均附有病例数 (N)、结果患病率和 AUC 的 95% 置信区间。标有“X”的点表示 ETHOS 模型中用于分类决策的特定阈值。还提供了精确召回 (PR) 曲线下的面积，并且 PR 曲线在补充材料中给出。现有研究的 AUC 代表了文献中确定的最佳算法的性能，并在文本中提供了参考文献。

为评估 ETHOS 的性能提供了有价值的基准。此外，我们对 ETHOS 与内部开发的专门算法进行了直接比较分析，这些发现在补充材料中进行了详细说明。

我们进行了一项分析，重点关注住院患者和 ICU 死亡率的风险评估，在患者入院和 ICU 的各个时间点进行计算。测试集包括 43,309 例住院患者（死亡率为 2.0%）和 7483 例住院患者（死亡率为 7.0%）。 ETHOS 模型表现出稳健的性能，医院死亡率的 AUC 为 0.921（95% CI：0.908-0.931），ICU 死亡率的 AUC 为 0.927（95% CI：0.914-0.938）。相比之下，在 ICU 死亡风险预测领域，我们的文献综述中确定的最高性能是 Pang 等人[17]使用 XGBoost 模型报告的 AUC 为 0.918（95% CI：0.915-0.922）。在较低端，Chen 等人[18] 报告的 AUC 为 0.642 ± 0.101。在由 3,324 名死亡率为 10.8% 的脓毒症患者组成的测试集中的特定亚组中，ETHOS 对 ICU 死亡率的预测显示 AUC 为 0.889（95% CI：0.870-0.906），这比 Pan 等人的一项研究中获得的结果更好。[19]（该研究估计了 ICU）使用 SOFA 和其他功能提高成年脓毒症患者的死亡率，AUC 为 0.762 ± 0.006。我们还评估了留在 ICU 至少 24 小时的患者 ICU 死亡率估计任务的性能，其中我们获得的 AUC 为 0.928（95% CI：0.916-0.939）。

此外，ETHOS 估计 ICU 住院时间 (LoS) 的平均绝对误差 (MAE) 为 2.262 天（95% CI：2.161–2.355 天）。这些结果与 [18] 的结果相似，后者报告的 MAE 为 2.42 ± 0.10 天。 ICU LoS 预测和死亡风险，强调了 ETHOS 在多个关键医疗保健指标上的竞争性零样本性能。

对于 ICU 再入院任务，ETHOS 的 AUC 为 0.807（95% CI：0.786-0.827），略小于使用知识图嵌入[20]获得的 AUC 0.82，并且高于使用基于 MIMIC-III 数据的 LSTM 获得的 AUC 0.791（95% CI，0.782-0.800）[21]。此外，我们将我们的方法应用于一项患病率相对较低的任务，特别关注我们测试集中入住 ICU 的 174 例出血患者。 ETHOS 对再入院的预测得出的 AUC 为 0.667（95% CI：0.402-0.839），与

AUC of 0.736 (95% CI: 0.668-0.801) achieved by previous studies[22] using LightGBM. For hospital readmission, ETHOS achieved an AUC of 0.749 (95% CI: 0.743-0.755), lower than the AUC of 0.791 [0.766-0.816] obtained by Tang et al.[23]. It's important to recognize that although MIMIC offers a wealth of data on acute care, it might not encompass all the subtleties necessary for readmission research, including comprehensive post-discharge outcomes or data on readmissions to various hospitals. Consequently, the accuracy of results for tasks related to readmission may be limited, regardless of the method employed.

We explored the task of predicting the first-day SOFA score at the time of admission. Given that the SOFA score is a critical indicator of survival, particularly in sepsis[6,24], this prediction can serve as a valuable indirect prognostic marker of ICU patient health status. We achieved a SOFA score estimation with an MAE of 1.502 (95% CI: 1.475-1.534). To our knowledge, no prior literature predicts first-day SOFA at the time of admission.

For the DRG assignment, we observed a top-1 (out of 771 classes) accuracy rate of 84.8% (95% CI: 84.4%–85.2%) in 28,932 hospitalizations using our methodology, a significant improvement over the 52% reported by Wang et al.[13], who explored DRG estimation using LLMs from discharge notes. This marked enhancement in performance can be attributed to the comprehensive nature of ETHOS, which incorporates a wide array of clinical events leading up to discharge within the PHT. In contrast, the approach taken by Wang et al.[13] relies solely on discharge notes, which may not encompass the breadth of information captured by PHT, thus potentially explaining the disparity in accuracy rates.

We want to reiterate an important point: all comparisons presented in this section are made between ETHOS, trained indiscriminately on the entire test population and task-specific algorithms developed using much smaller MIMIC data subsets obtained after data curation. In addition to the results in this section, in supplementary materials, we benchmark the performance of ETHOS against XGBoost[25], recurrent neural networks, and logistic regression.

Discussion

This work introduces an innovative approach to developing a Foundation Model for medical data derived from EMRs, designed to execute zero-shot inferences across a diverse range of tasks. Our model generates interpretable, causally forecasted future patient health timelines. By "causal," we mean that predictions are made solely based on information that occurred in the past. We applied and evaluated this model using the MIMIC-IV EMR datasets, comparing its performance with the results of methods published in the literature for the same tasks. Our objective, however, was not merely to surpass the performance of these specialized SOTA implementations. Instead, we aimed to demonstrate that ETHOS, a single foundation model trained just once with zero-shot derived inference, can achieve performance levels comparable to that of multiple models optimized for various tasks. This underscores the potential of ETHOS to streamline the application of AI in healthcare by leveraging a single unified model development architecture and set of methods for multiple prediction tasks, thereby greatly enhancing medical data model development efficiency and scalability.

The application of patient timelines for generating insights has been established in existing research[12,14–16,26], as has the implementation of foundation models[7]. Our methodology sets itself apart by integrating a zero-shot capability, obviating the need for additional training beyond the initial model. The design of ETHOS accommodates various approaches to inference, including few-shot predictions, although this necessitates fine-tuning for specific downstream tasks. Notably, the zero-shot prediction methodology introduces capabilities absent in few-shot prediction. To forecast future outcomes, ETHOS generates multiple health timelines representing possible future scenarios. This functionality exploits the model's capacity to explore and evaluate potential future events, thereby potentially estimating uncertainties. Future work will undoubtedly concentrate on refining this aspect of ETHOS. Moreover, ETHOS is specifically engineered to produce causal predictions in the form of future timelines, ensuring they are inherently comprehensible to human users. This is achieved through a novel tokenization process for medical data, a distinctive feature of our work.

The generation of multiple scenarios using the zero-shot approach places significant demands on time and computational resources. We estimated the inference time based on the average duration required to generate 1000 tokens. On a single Nvidia A100 GPU, this process took ~ 15 s. Given that computations are executed in batches and are thus highly parallelizable, we anticipate that in a potential production environment, response times could vary between 1 and 30 s. This variation is contingent upon the complexity of the downstream task at hand.

Another highly distinctive capability of ETHOS is the potential to generate individualized care-integrated PHT-based projected healthcare expenditures. This capability is exemplified through the prediction of Diagnosis-Related Group (DRG) codes but is not limited to this application. Specifically, ETHOS can model future PHTs at critical decision-making junctures in patient care. For instance, ETHOS can model outcomes for administering either drug A or B, considering the patient's unique conditions (such as sex, age, race, gender, income, etc) to determine which path might yield better clinical and cost outcomes. In this regard, ETHOS has the potential to revolutionize medical decision-analytic modeling science by incorporating a level of personalization previously unavailable in conventional decision-analytic models. This has the potential to enhance clinical decision-making and incorporate individualized real-time quantitatively robust value-based care policies into clinical care. This is a potentially transformative change, radically unlike current evidence-based medicine practices, which rely on high-quality data obtained from and averaged across patient populations[10,27,28]

In designing ETHOS, we have considered explainability, fairness, and transparency. These are vital aspects of our ongoing research. In future work, we plan to implement and test advanced visualization attention layers of the transformer[29] to gain insights into the model's reasoning process. Additionally, a dedicated interface for decision-making is envisaged further to enhance the usability of ETHOS in clinical settings.

Envisioning the development of a robust AI method that offers fully personalized advice on a wide range of medical questions necessitates learning from an extensive dataset of patients. Such a model must assimilate as much data as possible and be adaptable to a vast array of medical tasks. ETHOS represents a significant stride in this direction. Built on a transformer architecture, it is inherently scalable and, as a zero-shot learner, is versatile enough to address numerous key medical prediction tasks without task-specific training. Currently, ETHOS does not incorporate various types of critical information, including clinical and discharge notes, medical imaging and pathology images, genetic data, socioeconomic factors, lifestyle considerations, and monitoring signals. Nonetheless, the conceptual framework for incorporating these diverse data types is relatively straightforward. This can be done by leveraging the encoder and cross-attention mechanisms inherent in the transformer architecture; we anticipate the potential for integrating a nearly limitless amount of information during training. This expansion of ETHOS's capabilities forms the cornerstone of our future work, promising to enhance its applicability and efficacy in personalized medical advice and diagnostics.

We aim to modify further and train ETHOS to apply it across diverse data sources. This capability is currently hindered by variations in data collection methodologies, disparities in data quality, and the presence or absence of certain data types across different sources. Additionally, non-overlapping populations present significant challenges, rendering ETHOS not yet generalizable. To mitigate some of these compatibility issues, we propose the development of a universal tokenization format[30]. While this approach may resolve certain discrepancies, it does not address all underlying compatibility concerns. The ultimate solution, we believe, lies in a system capable of transforming tokenized data from one healthcare system to another, akin to text translation between languages. Specifically, for ETHOS, this would mean converting the patient journey, as encapsulated by the PHT, from one system's format to another. This conversion would not only facilitate a consistent and unified representation of patient histories across different systems but also offer insights into the operational nuances of these systems. Pursuing such a translation strategy represents a vital

先前使用 LightGBM 的研究[22]实现了 0.736 的 AUC（95% CI：0.668-0.801）。对于再入院，ETHOS 的 AUC 为 0.749（95% CI：0.743-0.755），低于 Tang 等人获得的 0.791 [0.766-0.816] 的 AUC。[23]。重要的是要认识到，尽管 MIMIC 提供了大量有关急症护理的数据，但它可能无法涵盖再入院研究所需的所有微妙之处，包括全面的出院后结果或各医院再入院的数据。因此，无论采用何种方法，与再入院相关的任务结果的准确性可能会受到限制。

我们探索了预测入院时第一天 SOFA 分数的任务。鉴于 SOFA 评分是生存的关键指标，特别是脓毒症[6,24]，因此该预测可以作为 ICU 患者健康状况的有价值的间接预后标志。我们实现了 MAE 为 1.502 的 SOFA 评分估计（95% CI：1.475-1.534）。据我们所知，之前没有文献预测入院时的第一天 SOFA。

对于 DRG 分配，我们使用我们的方法在 28,932 例住院治疗中观察到 top-1（771 个类别中）准确率为 84.8%（95% CI：84.4%–85.2%），比 Wang 等人报告的 52% 有显着提高。[13]，他们使用出院记录中的 LLM 探索了 DRG 估计。性能的显着提高可归因于 ETHOS 的综合性质，其中包含导致 PHT 内出院的各种临床事件。相比之下，Wang 等人[13]采取的方法仅依赖于出院记录，这可能不包含 PHT 捕获的信息的广度，因此可能解释了准确率的差异。

我们想重申一个重要的观点：本节中提出的所有比较都是在对整个测试群体进行不加区别训练的 ETHOS 和使用数据管理后获得的更小的 MIMIC 数据子集开发的特定于任务的算法之间进行的。除了本节中的结果之外，在补充材料中，我们还针对 XGBoost[25]、循环神经网络和逻辑回归对 ETHOS 的性能进行了基准测试。

讨论

这项工作引入了一种创新方法，用于为源自 EMR 的医疗数据开发基础模型，旨在跨各种任务执行零样本推理。我们的模型生成可解释的、因果预测的未来患者健康时间表。我们所说的“因果”是指仅根据过去发生的信息做出预测。我们使用 MIMIC-IV EMR 数据集应用和评估了该模型，将其性能与文献中针对相同任务发布的方法的结果进行了比较。然而，我们的目标不仅仅是超越这些专门的 SOTA 实现的性能。相反，我们的目的是证明 ETHOS（通过零样本派生推理仅训练一次的单一基础模型）可以达到与针对各种任务优化的多个模型相当的性能水平。这凸显了 ETHOS 通过利用单一统一模型开发架构和一组用于多个预测任务的方法来简化人工智能在医疗保健中的应用的潜力，从而大大提高医疗数据模型开发效率和可扩展性。

现有研究[12,14–16,26]已经确立了应用患者时间线来产生见解，以及基础模型的实施[7]。我们的方法通过集成零样本功能而脱颖而出，无需在初始模型之外进行额外的训练。 ETHOS 的设计适应了各种推理方法，包括小样本预测，尽管这需要针对特定下游任务进行微调。值得注意的是，零样本预测方法引入了少样本预测中所缺乏的功能。为了预测未来的结果，ETHOS 生成多个代表未来可能情况的健康时间表。此功能利用模型探索和评估潜在未来事件的能力，从而潜在地估计不确定性。未来的工作无疑将集中于完善 ETHOS 的这一方面。此外，ETHOS 经过专门设计，可以以未来时间线的形式产生因果预测，确保人类用户本质上可以理解它们。这是通过一种新颖的医疗数据标记化流程来实现的，这是我们工作的一个显着特征。

使用零样本方法生成多个场景对时间和计算资源提出了很高的要求。我们根据生成 1000 个令牌所需的平均持续时间来估计推理时间。在单个 Nvidia A100 GPU 上，此过程花费了 ~ 15 秒。鉴于计算是批量执行的，因此具有高度并行性，我们预计在潜在的生产环境中，响应时间可能在 1 到 30 秒之间变化。这种变化取决于当前下游任务的复杂性。

ETHOS 的另一个非常独特的功能是能够生成基于 PHT 的个性化护理集成预计医疗保健支出。此功能通过诊断相关组 (DRG) 代码的预测来举例说明，但不限于此应用。具体来说，ETHOS 可以在患者护理的关键决策时刻模拟未来的 PHT。例如，ETHOS 可以对服用药物 A 或 B 的结果进行建模，考虑患者的独特条件（例如性别、年龄、种族、性别、收入等），以确定哪种途径可能会产生更好的临床和成本结果。在这方面，ETHOS 有潜力通过整合传统决策分析模型中以前无法实现的个性化水平来彻底改变医疗决策分析建模科学。这有可能增强临床决策并将个性化的实时定量稳健的基于价值的护理政策纳入临床护理中。这是一个潜在的变革，与当前的循证医学实践截然不同，后者依赖于从患者群体中获得的高质量数据并对其进行平均[10,27,28]

在设计 ETHOS 时，我们考虑了可解释性、公平性和透明度。这些是我们正在进行的研究的重要方面。在未来的工作中，我们计划实现和测试 Transformer[29] 的高级可视化注意力层，以深入了解模型的推理过程。此外，还设想了一个用于决策的专用界面，以进一步增强 ETHOS 在临床环境中的可用性。

设想开发一种强大的人工智能方法，为广泛的医疗问题提供完全个性化的建议，需要从广泛的患者数据集中学习。这样的模型必须吸收尽可能多的数据并适应大量的医疗任务。 ETHOS 代表了朝这个方向迈出的一大步。它建立在 Transformer 架构之上，具有本质上的可扩展性，并且作为零样本学习器，它具有足够的通用性，无需进行特定任务的训练即可解决众多关键的医学预测任务。目前，ETHOS 尚未纳入各种类型的关键信息，包括临床和出院记录、医学成像和病理图像、遗传数据、社会经济因素、生活方式考虑因素和监测信号。尽管如此，整合这些不同数据类型的概念框架相对简单。这可以通过利用 Transformer 架构中固有的编码器和交叉注意力机制来完成；我们预计在培训期间整合几乎无限量的信息的潜力。 ETHOS 功能的扩展构成了我们未来工作的基石，有望增强其在个性化医疗建议和诊断方面的适用性和有效性。

我们的目标是进一步修改并训练 ETHOS 以将其应用于不同的数据源。目前，这种能力受到数据收集方法的差异、数据质量的差异以及不同来源中是否存在某些数据类型的阻碍。此外，非重叠人群带来了重大挑战，使得 ETHOS 尚未推广。为了缓解其中一些兼容性问题，我们建议开发通用标记化格式[30]。虽然这种方法可以解决某些差异，但它并不能解决所有潜在的兼容性问题。我们相信，最终的解决方案在于一种能够将标记化数据从一个医疗系统转换到另一个医疗系统的系统，类似于语言之间的文本翻译。具体来说，对于 ETHOS 来说，这意味着将 PHT 封装的患者旅程从一种系统的格式转换为另一种系统的格式。这种转换不仅有助于跨不同系统对患者病史进行一致和统一的表示，而且还可以深入了解这些系统的操作细微差别。追求这样的翻译策略代表着至关重要的

direction for our future research endeavors, alongside evaluating the methodologies introduced in this paper through analysis of prospectively collected data.

ETHOS and LLMs such as GPT-4o, Claude 3 Opus, and Gemini 1.5 Ultra, although built upon similar AI principles, serve different purposes and exhibit distinct capabilities. ETHOS is specifically designed to predict fPHTs through explicit modeling of quantitative values and temporal sequences. This approach allows ETHOS to leverage structured patient data to generate predictions. In contrast, LLMs are general-purpose models optimized for tasks involving knowledge integration, reasoning, and interactive conversation. They do not explicitly model quantitative values and time sequences, which are important for accurate clinical decision support. Studies, such as those by Hager et al.[31] and Wang and Zhao[32], highlight the limitations of LLMs in handling temporal information and decision support tasks, emphasizing the potential need for specialized models like ETHOS. There is a potential of ETHOS to be used in conjunction with LLMs through retrieval-augmented generation (RAG) mechanisms, offering a promising direction for future AI applications in healthcare. In supplementary material, we present a comparison in predictive performance of ETHOS and LLM (GPT-4o). Furthermore, while LLMs excel in processing vast amounts of unstructured text, their computational performance in generating detailed and contextually accurate patient predictions remains suboptimal compared to ETHOS because of efficient representation of information in ETHOS tokenized PHTs.

This work has limitations. We utilized the MIMIC dataset, which may be cleaner than many routine clinical datasets. Performance and usability should be tested prospectively in diverse datasets and in real-time. The transformer model in the current version of ETHOS is relatively simple and uses only 2048 PHT tokens for predictions. When token density per time is large, this may not contain sufficient information for optimal performance. Mitigation of the limitation is expected with additional computational infrastructure.

In conclusion, ETHOS presents a promising approach to deriving insights from massive clinical datasets without labor-intensive labeling or distinct model creation for each prediction task. This approach has the potential to significantly lower the costs and complexities associated with AI model development, thereby accelerating the development and implementation of healthcare AI.

Methods

Data

In this study, the Medical Information Mart for Intensive Care (MIMIC-IV) database served as a data source, providing a rich and comprehensive collection of de-identified health-related information[4]. Managed collaboratively by the Massachusetts Institute of Technology (MIT), Beth Israel Deaconess Medical Center (BIDMC), and Philips Healthcare, MIMIC-IV encompasses detailed records for >200,000 patients who were admitted to hospital and critical care units at BIDMC in Boston, Massachusetts, between 2008 and 2019. The following tables from the MIMIC-IV were used: (1) *Patients*, which contains static information about the patients, such as gender, date of birth, and date of death; (2) *Admissions*, which holds information about patient admissions to the hospital, including admission and discharge times, as well as information related to the hospital stay; (3) *Icustays*, which is specifically related to intensive care unit (ICU) stays, including the timings and type of ICU; (4) *Labevents*, which contains laboratory test results for patients. We used the 200 most frequent tests covering 95% of tests completed; (5) *Prescriptions*, which holds information on medications prescribed to patients during their stay, with each drug converted to ATC code[33] We converted GSN codes in MIMIC-IV to ATC codes using conversion tables[26]; (6) *Procedures* which contains information about procedures performed on patients, coded using ICD10-PCS codes; (7) *Diagnoses* which contains diagnostic information, typically coded using ICD10-CM codes. We converted ICD9 to ICD10-CM if needed using conversion table[34]; (8) *Emar*, which holds information related to the documentation and administration of medications to patients; (9) *Omr* with information about measurements taken from a patient, such as blood pressure or BMI; (10) *Services* with information about the clinical service under which a patient is managed during their hospital stay; (11) *drgcodes* DRG codes which are a classification system used in the healthcare industry to categorize hospital cases into groups that are expected to have similar hospital resource use; (12) SOFA, taken from the derived tables in MIMIC. The remaining tables were not used in the current ETHOS implementation as they will require additional processing. For example, clinical notes require natural language processing to be converted to meaningful tokenized information.

Patient health timelines (PHTs), tokenization

The core concept behind ETHOS is the Patient Health Timeline (PHT), as depicted in Fig. 1. The fundamental component of the PHT is the token, which represents a distinct unit of information occurring within the patient's health timeline. To construct the PHT, we gathered all pertinent data from tables 1 to 12 of the MIMIC-IV database, as detailed in the *Data* section. We arranged this data chronologically based on timestamps, as shown in Fig. 4a, into a chronological sequence of health-related events for each patient. These events were timestamped with a floating-point number in 64-bit precision to denote the patient's age at the time of occurrence of the event. Subsequently, events from the MIMIC-IV tables were converted into tokens. Each event was represented by 1 to 7 tokens to encapsulate information about the event, as illustrated in Supplementary Fig. 5a. We crafted this encoding process to ensure each token conveys specific, meaningful information, with examples in Supplementary Fig. 5c–f. A comprehensive list of token encodings within the PHT is available in the supplementary material. The final step of tokenization involved the insertion of time-interval tokens to represent the intervals between events, depicted in Fig. 2c. We employed 13 different time-interval tokens to represent the intervals. No interval token was inserted if the duration between tokens was <5 min. Typically, a single time-interval token was placed between other types of tokens unless the interval exceeded 1 year. In such cases, multiple 6-month tokens were used to approximate the actual interval. For example, an interval of 1.4 years was represented by three 6-month tokens, while four 6-month tokens represented 1.76 years. One interval-tokens were inserted the exact time of events was dropped from PHTs.

The patient's age and the commencement date of the PHT were represented using the same token set. We used 20 distinct tokens to denote age intervals such as 0–5 years, 5–10 years, and so forth. For instance, to encode information about a 46 year-old patient with PHT beginning in 1982, we inserted a "45–50 years" token at the 4th position in the PHT. To signify the year 1982, we used a "15–20 years" token at the 5th position of the PHT, considering 1970 as the baseline year. We emphasize that age and the commencement of the PHT are encoded in 5 year intervals, given that health status typically does not undergo rapid changes with age, making finer granularity unnecessary. However, we plan to scrutinize these assumptions in subsequent research. The token denoting the commencement of the PHT delineates the temporal context of the medical data—identifying whether it corresponds to earlier medical practices (e.g., 1990s), contemporary practices, or periods in between. Using tokens with a precision of 5 years is done under the premise that technological and methodological progress within the medical field does not advance at a pace that justifies the necessity for time intervals more granular than 5 year spans. Pertinent to the MIMIC dataset, the obfuscation of actual dates through uniform random adjustments for each patient—a measure implemented to safeguard privacy—compromises the utility of this temporal information for ETHOS, as it obscures the precise date of the start of PHT. However, the absence of precise reference dates is less critical, given that the entire dataset was collected over a relatively brief period, from 2008 to 2019[5].

As mentioned previously, token locations within the timeline are contingent upon the temporal occurrence of events. Nonetheless, certain data elements are temporally invariant, or at least presented as such within the MIMIC-IV database. In our implementation, we designate six static tokens to encapsulate the information encoded in these static data elements.

我们未来研究工作的方向，同时通过分析前瞻性收集的数据来评估本文中介绍的方法。

ETHOS 和 LLM（例如 GPT-4o、Claude 3 Opus 和 Gemini 1.5 Ultra）虽然建立在类似的人工智能原理之上，但服务于不同的目的并表现出不同的功能。 ETHOS 专门设计用于通过定量值和时间序列的显式建模来预测 fPHT。这种方法允许 ETHOS 利用结构化患者数据来生成预测。相比之下，法学硕士是针对涉及知识整合、推理和交互式对话的任务进行优化的通用模型。它们没有明确地模拟定量值和时间序列，这对于准确的临床决策支持很重要。 Hager 等人[31]、Wang 和Zhao[32] 等人的研究强调了法学硕士在处理时间信息和决策支持任务方面的局限性，强调了对 ETHOS 等专门模型的潜在需求。 ETHOS 有潜力通过检索增强生成（RAG）机制与法学硕士结合使用，为未来人工智能在医疗保健领域的应用提供了一个有希望的方向。在补充材料中，我们对 ETHOS 和 LLM (GPT-4o) 的预测性能进行了比较。此外，虽然法学硕士擅长处理大量非结构化文本，但与 ETHOS 相比，它们在生成详细且上下文准确的患者预测方面的计算性能仍然不够理想，因为 ETHOS 标记化 PHT 中的信息可以有效表示。

这项工作有局限性。我们使用了 MIMIC 数据集，它可能比许多常规临床数据集更干净。性能和可用性应该在不同的数据集中实时进行前瞻性测试。当前版本 ETHOS 中的 Transformer 模型相对简单，仅使用 2048 个 PHT 代币进行预测。当每次的令牌密度很大时，这可能不包含获得最佳性能的足够信息。预计通过额外的计算基础设施可以缓解该限制。

总之，ETHOS 提出了一种有前途的方法，可以从大量临床数据集中获取见解，而无需为每个预测任务进行劳动密集型标记或创建不同的模型。这种方法有可能显着降低与人工智能模型开发相关的成本和复杂性，从而加速医疗保健人工智能的开发和实施。

方法

数据

在这项研究中，重症监护医疗信息集市（MIMIC-IV）数据库作为数据源，提供了丰富且全面的去识别化健康相关信息的集合[4]。 MIMIC-IV 由麻省理工学院 (MIT)、贝斯以色列女执事医疗中心 (BIDMC) 和飞利浦医疗保健公司合作管理，包含 2008 年至 2019 年间在马萨诸塞州波士顿 BIDMC 住院和重症监护病房的 200,000 名患者的详细记录。使用了 MIMIC-IV 中的以下表格：(1) 患者，其中包含患者的静态信息，例如性别、出生日期和死亡日期； （2）入院信息，保存患者入院信息，包括入院、出院时间以及住院相关信息； (3) Icustays，特别与重症监护病房 (ICU) 住院相关，包括 ICU 的时间和类型； (4) Labevents，其中包含患者的实验室测试结果。我们使用了 200 个最常见的测试，覆盖了已完成测试的 95%； (5) 处方，保存患者住院期间开出的药物信息，每种药物均转换为 ATC 代码[33] 我们使用转换表[26]将 MIMIC-IV 中的 GSN 代码转换为 ATC 代码； (6) 程序，其中包含有关对患者进行的程序的信息，使用 ICD10-PCS 代码进行编码； (7) 包含诊断信息的诊断，通常使用 ICD10-CM 代码进行编码。如果需要，我们使用转换表[34]将ICD9转换为ICD10-CM； (8) Emar，保存与患者用药记录和管理相关的信息； (9) 欧姆尔与有关患者测量结果的信息，例如血压或体重指数； (10) 包含患者住院期间接受管理的临床服务信息的服务； (11) drgcodes DRG 代码，这是医疗行业中使用的分类系统，用于将医院病例分类为预计具有类似医院资源使用的组； (12) SOFA，取自 MIMIC 中的派生表。其余表未在当前 ETHOS 实施中使用，因为它们需要额外处理。例如，临床记录需要自然语言处理才能转换为有意义的标记化信息。

患者健康时间表 (PHT)、标记化

ETHOS 背后的核心概念是患者健康时间线 (PHT)，如图 1 所示。PHT 的基本组成部分是代币，它代表患者健康时间线内发生的独特信息单元。为了构建 PHT，我们从 MIMIC-IV 数据库的表 1 到表 12 收集了所有相关数据，如数据部分所述。我们根据时间戳按时间顺序排列这些数据，如图 4a 所示，按照每个患者的健康相关事件的时间顺序排列。这些事件带有 64 位精度的浮点数时间戳，以表示事件发生时患者的年龄。随后，MIMIC-IV 表中的事件被转换为令牌。每个事件都由 1 到 7 个标记来表示，以封装有关该事件的信息，如补充图 5a 所示。我们精心设计了这个编码过程，以确保每个标记都传达特定的、有意义的信息，示例见补充图 5c-f。补充材料中提供了 PHT 中令牌编码的完整列表。标记化的最后一步涉及插入时间间隔标记来表示事件之间的间隔，如图 2c 所示。我们使用 13 种不同的时间间隔标记来表示间隔。如果令牌之间的持续时间为 <5 分钟，则不会插入间隔令牌。通常，除非间隔超过 1 年，否则会将单个时间间隔令牌放置在其他类型的令牌之间。在这种情况下，使用多个 6 个月的令牌来近似实际间隔。例如，1.4 年的间隔由 3 个 6 个月的代币表示，而 4 个 6 个月的代币则表示 1.76 年。插入一个间隔标记，事件的确切时间从 PHT 中删除。

患者的年龄和 PHT 的开始日期使用相同的标记集表示。我们使用 20 个不同的标记来表示年龄区间，例如 0-5 岁、5-10 岁等。例如，为了对 1982 年开始患有 PHT 的 46 岁患者的信息进行编码，我们在 PHT 的第 4 个位置插入了“45-50 年”标记。为了表示 1982 年，我们在 PHT 的第 5 位使用了“15-20 年”标记，并将 1970 年视为基准年。我们强调，年龄和 PHT 的开始时间以 5 年为间隔进行编码，因为健康状况通常不会随年龄发生快速变化，因此不需要更细的粒度。然而，我们计划在后续研究中仔细审查这些假设。表示 PHT 开始的标记描绘了医疗数据的时间背景——识别它是否对应于早期的医疗实践（例如 20 世纪 90 年代）、当代实践或之间的时期。使用精度为 5 年的代币是在以下前提下进行的：医疗领域内的技术和方法进步的速度不足以证明需要比 5 年跨度更细粒度的时间间隔。与 MIMIC 数据集相关，通过对每位患者进行统一随机调整（一种保护隐私的措施）来混淆实际日期，会损害 ETHOS 的时间信息的实用性，因为它模糊了 PHT 开始的精确日期。然而，鉴于整个数据集是在 2008 年至 2019 年相对较短的时间内收集的，因此缺乏精确的参考日期并不那么重要[5]。

如前所述，时间线内的令牌位置取决于事件的时间发生。尽管如此，某些数据元素在时间上是不变的，或者至少在 MIMIC-IV 数据库中如此呈现。在我们的实现中，我们指定了六个静态标记来封装这些静态数据元素中编码的信息。

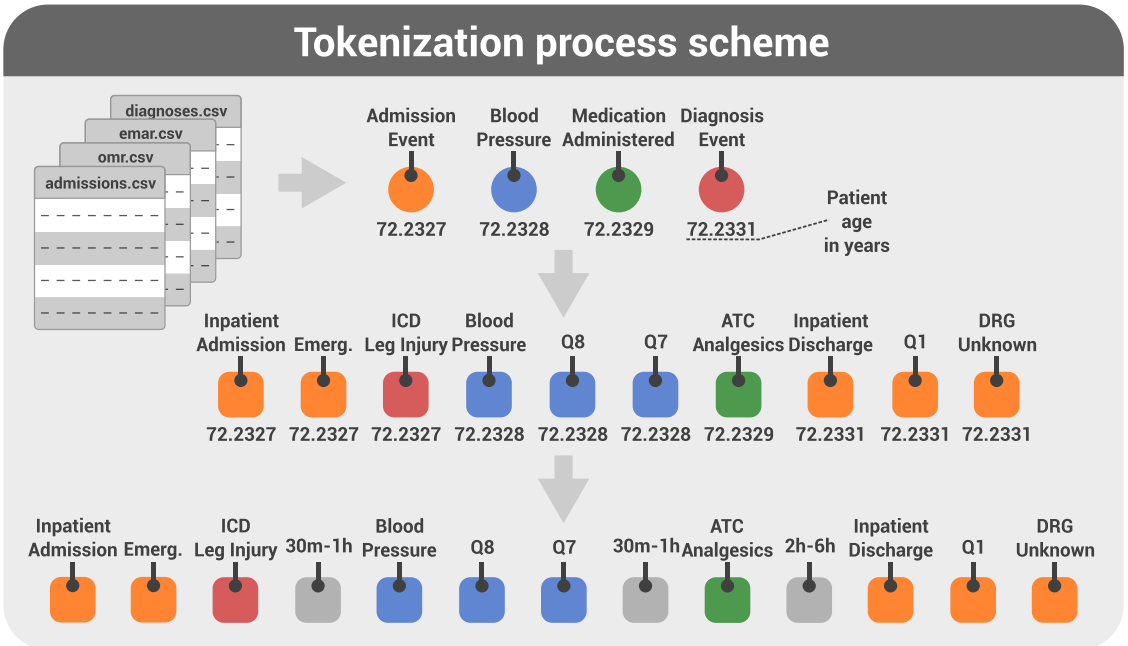

Fig. 4 | Stages of PHT construction and tokenization in ETHOS. The process begins with assembling a chronological list of events from MIMIC-IV tables, Each entry on the list is time stamped with 64-bit real value only 6 significant digits show for clarity, indicating the patient's age at which the event occurred. Subsequently, list elements are transformed into tokens using ETHOS tokenization scheme. Based on the event's nature, one event can be translated into 1 up to 7 tokens. Each token derived from the same event shares its timestamp. The final step involves representing time gaps between events by inserting time-interval tokens. If the time difference between events is <5 min—the minimum value represented by the token for the shortest time interval—no token is added. After adding interval-tokens, timestamps are stripped from the timeline.

Although, in reality, some of these variables may change over time, they are represented as invariable constants in the MIMIC database. We encoded this information in the six static tokens exactly as recorded in the MIMIC dataset. These include gender, marital status, race, body mass index (BMI), birth date, and the start date of the timeline. While PHTs have the potential to extend to hundreds of thousands of tokens, our current methodology utilizes a maximum of 2048 subsequent tokens within the transformer model context, as elaborated in the "Methods: ETHOS Training" section. To accommodate invariant data, we substitute the initial six tokens of the 2048-token context with static information tokens, where the sixth token demarcates the temporal juncture of the seventh token, which is the first token of the actual timeline. Although the transformer architecture inherently facilitates the inclusion of static data via its encoder component and cross attention module[3], we opted for a more streamlined approach as described, deferring the integration of an encoder implementation to future endeavors where more substantial time-invariant data like genetics is used.

Medical encounters yield a plethora of numerical data. We employ a quantile-based tokenization strategy to process continuous numerical values, such as blood pressure readings or cholesterol levels. Specifically, all numerical values are transformed into integers representing the quantile to which each value corresponds. Quantile ranges were determined using the training dataset, where histograms of all numerical values were generated and subsequently divided into quantiles. We chose to utilize ten quantiles, a decision aimed at striking a balance between the need for precise representation of numerical data and the clinical reality that significant changes in health indicators often manifest as relatively large variations, such as shifts of 10 or 20 percent (Supplementary Fig. 4). This rationale underpins our selection of ten quantiles for tokenization.

In our study, Diagnosis-Related Group (DRG) codes for each inpatient stay were utilized, despite the absence of assigned times when they were created in the MIMIC tables. Given that a DRG code is assigned after or during discharge, we positioned it after a trio of tokens representing discharge-related information: the discharge token, a quantile token indicating the length of the hospital stay, and a token specifying the discharge destination (e.g., home). Additionally, we incorporated data from MIMIC regarding the initial SOFA score for ICU patients, placing this token after the patient's admission-to-the-ICU token, along with a token denoting the ICU type. Given that the SOFA score in the dataset ranges from 0 to 23 (with the score of 24 never appearing), we uniformly map scores from 0–23 across 1–10 quantiles. Consequently, in quantile Q1, SOFA scores of 0, 1, and 2 (average of 1) are included, while quantile Q2 encompasses SOFA scores of 3 and 4 (average of 3.5), and this pattern continues accordingly.

ETHOS operates as a causal network. It relies solely on information available up to the time being considered in making predictions. Consequently, to ensure causality, actual values of DRG codes and SOFA scores are not employed during inference; instead, predictions of these values are used. This principle ensures that future-obtained information does not influence the prediction of yet-to-occur events. In essence, if tokens are integrated into the timeline based on their approximate occurrence time, their actual values must not be utilized for inference purposes, or they are placed in the timeline far in the future to ensure they are inserted after they occurred.

For the tokenization of drugs, whether administered or prescribed, we utilized the ATC classification system due to its hierarchical, tree-like structure (Supplementary Fig. 1). Each ATC code, comprising up to seven characters, was encoded using up to three sequential tokens: the first token for the initial three characters, the second for the subsequent character, and the third optional token, for the remaining suffix. Similarly, ICD-10-CM codes were encoded with three tokens: the first representing the first three characters of the code, the next two by the second token, and the final token capturing the code's remaining suffix. For ICD-10-PCS codes, each character in the seven-character code was represented by a distinct token. The rationale behind such tokenization is that the initial characters in those coding schemes denote specific classes of drugs and diseases or procedures, which are interpretable and have distinct meanings which we anticipated to be important for the network's self-attention mechanisms. Looking ahead, our approach, which assigns well-defined meanings to each token, will be crucial for refining attention mechanisms and enhancing the model's

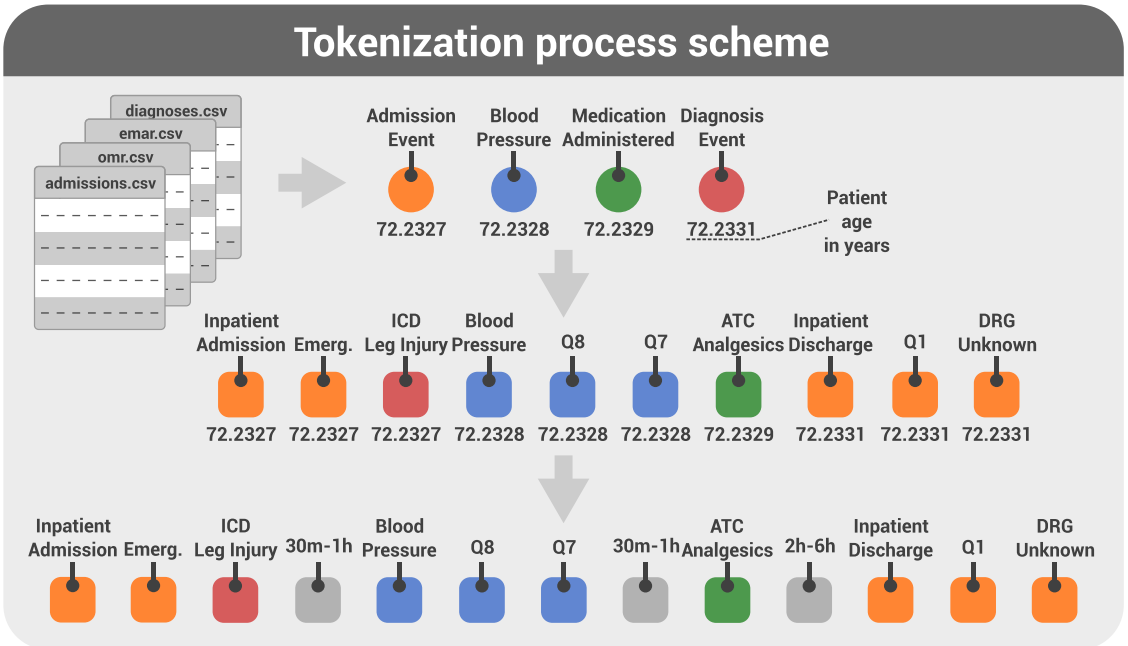

图 4 | ETHOS 中 PHT 构建和代币化的阶段。该过程首先从 MIMIC-IV 表中组装按时间顺序排列的事件列表，列表中的每个条目都带有 64 位实际值的时间戳，为了清晰起见，仅显示 6 位有效数字，表明事件发生时患者的年龄。随后，使用 ETHOS 标记化方案将列表元素转换为标记。基于根据事件的性质，一个事件可以转化为 1 至 7 个代币。每个令牌从同一事件派生的共享其时间戳。最后一步涉及通过插入时间间隔标记来表示事件之间的时间间隔。如果事件之间的时间差为 <5 分钟（最短时间间隔的令牌表示的最小值），则不添加令牌。添加间隔标记后，时间戳将从时间线中删除。

尽管实际上，其中一些变量可能会随着时间而变化，但它们在 MIMIC 数据库中表示为不变的常量。我们将此信息编码在六个静态标记中，与 MIMIC 数据集中记录的完全相同。其中包括性别、婚姻状况、种族、体重指数 (BMI)、出生日期和时间线的开始日期。虽然 PHT 有潜力扩展到数十万个代币，但我们当前的方法在 Transformer 模型上下文中最多使用 2048 个后续代币，如“方法：ETHOS 训练”部分中所述。为了适应不变的数据，我们用静态信息标记替换 2048 个标记上下文的最初六个标记，其中第六个标记划分了第七个标记的时间节点，第七个标记是实际时间线的第一个标记。尽管 Transformer 架构本质上促进了通过其编码器组件和交叉注意模块 [3] 包含静态数据，但我们选择了所描述的更简化的方法，将编码器实现的集成推迟到使用更大量的时不变数据（如遗传学）的未来工作中。

医疗经历会产生大量的数字数据。我们采用基于分位数的标记化策略来处理连续数值，例如血压读数或胆固醇水平。具体来说，所有数值都转换为表示每个值对应的分位数的整数。使用训练数据集确定分位数范围，其中生成所有数值的直方图，然后将其划分为分位数。我们选择使用十个分位数，这一决定旨在在数字数据精确表示的需求与临床现实之间取得平衡，即健康指标的显着变化通常表现为相对较大的变化，例如 10% 或 20% 的变化（补充图 4）。这一基本原理支撑了我们选择十个分位数进行标记化。

在我们的研究中，尽管在 MIMIC 表中创建时没有指定时间，但还是使用了每次住院的诊断相关组 (DRG) 代码。鉴于 DRG 代码是在出院后或出院期间分配的，我们将其放置在代表出院相关信息的三个标记之后：出院标记、指示住院时间长度的分位数标记以及指定出院的标记目的地（例如，家）。此外，我们还整合了来自 MIMIC 的有关 ICU 患者初始 SOFA 评分的数据，将该标记放置在患者入住 ICU 标记之后，以及表示 ICU 类型的标记。鉴于数据集中的 SOFA 分数范围为 0 到 23（分数 24 从未出现），我们将 0-23 的分数统一映射到 1-10 分位数。因此，在分位数 Q1 中，包含 SOFA 分数 0、1 和 2（平均值为 1），而分位数 Q2 包含 SOFA 分数 3 和 4（平均值为 3.5），并且该模式相应地继续下去。

ETHOS 作为一个因果网络运作。它仅依赖于做出预测时考虑的可用信息。因此，为了确保因果关系，在推理过程中不使用 DRG 代码和 SOFA 分数的实际值；相反，使用这些值的预测。这一原则确保未来获得的信息不会影响对尚未发生事件的预测。本质上，如果令牌根据其大致发生时间集成到时间线中，则它们的实际值不得用于推理目的，或者将它们放置在遥远的未来时间线中以确保它们在发生后插入。

对于药物的标记化，无论是管理还是处方，我们都利用了 ATC 分类系统，因为它具有分层的树状结构（补充图 1）。每个 ATC 代码最多包含七个字符，使用最多三个连续标记进行编码：第一个标记用于前三个字符，第二个标记用于后续字符，第三个可选标记用于其余后缀。类似地，ICD-10-CM 代码使用三个标记进行编码：第一个标记代表代码的前三个字符，接下来的两个标记代表第二个标记，最后一个标记捕获代码的剩余后缀。对于 ICD-10-PCS 代码，七字符代码中的每个字符都由一个不同的标记表示。这种标记化背后的基本原理是，这些编码方案中的初始字符表示特定类别的药物和疾病或程序，它们是可解释的并且具有不同的含义，我们预计这对于网络的自我注意机制很重要。展望未来，我们的方法为每个标记分配了明确定义的含义，这对于完善注意力机制和增强模型的能力至关重要

explainability. This method ensures that individual tokens contribute significantly to the interpretability of the network's outcomes. For more information on the tokenization process applied to MIMIC data in our analysis, as well as examples of Patient Health Timelines (PHTs), readers are directed to Supplementary Table 3 where we present real PHTs used in this work with annotations. A summary of all tokenized components of the MIMIC dataset is Supplementary Table 2.

ETHOS training

We employ a model inspired by the decoder architecture of the transformer[3], drawing parallels between tokenized text in Natural Language Processing (NLP) and our approach to tokenizing PHTs. We based our model development on Andrej Kapathy's implementation of GPT-2 (github.com/karpathy/nanoGPT). The design choice slightly varies from the original transformer paper, because instead of using fixed sinusoidal positional encodings, it utilizes learneable position embeddings that are added to the token embeddings at the stage where tokens are converted to their corresponding embeddings. The ETHOS model's training begins by synthesizing a dataset from existing patient records. Each patient's PHT is ended with a "End of timeline" token, and then they are concatenated, creating a single long sequence of tokens for the training. Similarly to generative LLM, ETHOS is trained to predict a single token based on the context of preceding ones. Given the large data scale and model complexity, this phase is resource-intensive similar to methods for training used for NLP transformers used in LLMs[3,35]. We estimated that the size of the network training task that we face with ETHOS is similar to GPT-2[8], and therefore we used the size of the transformer used in that network as a starting point (details on the hyperparameter search and choice can be seen in Supplementary Fig. 2). We made heuristic adjustments to the size of the network to optimize the value of the loss function. Further details on our training methodology of transformers are provided in Brown et al.[8] and for our implementation in supplementary material and full complete code published at https://github.com/ipolharvard/ethos-paper.

ETHOS inference

During inference, ETHOS functions analogous to a document completion tool in which word sequences instead of health-related events are sequenced into a PHT. The procedure begins with the patient's history recorded in their PHTs. The last 2048 tokens—or the entire PHT if it contains fewer than 2048 tokens—are used to initiate the inference in the current ETHOS implementation. ETHOS then generates one token at a time through the following steps: (1) generating an array of probabilities for all potential tokens, (2) stochastically selecting a new token based on these probabilities, (3) appending the new token to the sequence while removing the oldest one to maintain the context size at 2048 tokens, (4) go to 1. This generative sequence proceeds until it encounters predefined stopping conditions, which may include the appearance of a token showing the patient's death or the sum of time-interval tokens surpassing a certain threshold. Additional stopping criteria may be established. The stochastic nature of this method allows for the creation of multiple future PHTs (fPHTs). Multiple fPHTs are used to assess uncertainties as each of the fPHTs represents an alternative prediction of the future.

Evaluation of clinical outcomes and tasks using ETHOS

The experiments were chosen so the results can be compared to the work of others in terms of the estimation of inpatient mortality and readmission on MIMIC data. Patients in the MIMIC were randomly divided into training and testing groups, with splits of 90%/10% (Supplementary Table 1).

The chance of inpatient mortality was assessed at the time of admission for all inpatient stays for patients in the test set unless the discharge day was unknown. This was performed by the generative process that began with the admission token and ended upon generating a discharge or death token, repeating this cycle 20 times. The 'N', representing the number of times a death token was generated first, was divided by 20 to estimate the chance of inpatient mortality. Similarly, the likelihood of ICU mortality was computed for the MIMIC dataset, with an additional experiment conducted where predictions were made starting 24 h after ICU admission, rather than at the point of ICU admission. In the same simulation, the LOS in the ICU was estimated by aggregating the time-interval tokens generated in the simulated timeline until the discharge token appeared. Instances where the patient died in the ICU during the simulation were excluded from the LOS calculation. We opted for 20 repetitions, yielding 21 unique probability estimators, which were adequate for constructing robust Receiver Operating Characteristic (ROC) curves yielding excellent Gaussian fits (Fig. 3). Nevertheless, alternative repetition counts may also be employed.

To calculate the probability of 30 day inpatient readmission, the generation of fPHTs commenced at the discharge token from inpatient stays and ceased upon the appearance of either a new admission or death token or when the cumulative time tokens generated exceeded 30 days. The simulation was repeated 20 times. The probability of 30 day readmission was then derived as M/20, where 'M' is the count of terminations occurring because of patient new admission tokens across the 20 repetitions.

In our approach, tasks are accomplished by simulating future patient health timelines. Yet, ETHOS offers additional methods for deriving insights, two of which we illustrate here. For instance, in the construction of PHTs following each ICU admission, a sequence is created starting with a token that identifies the type of ICU, followed by a SOFA score token, and then by a Q token that signifies the actual SOFA score on the first day. We predict the SOFA score using SOFA Q node probabilities as generated by ETHOS and the mean SOFA score per quantile as assigned during tokenization (Fig. 5a).

The exact timing of the 1 day SOFA score assessment is not specified in the dataset, leading to a potential causality issue by inserting the SOFA score immediately after admission, as it relies on data acquired subsequently. During the model's training phase, ETHOS permits this apparent causality violation. However, such true values of 1 day SOFA scores, not available at the moment of ICU admission, are not used for simulating future timelines during inference to prevent causality violation during inference. Instead, these scores are predicted from prior information, as demonstrated in our study. This feature of ETHOS enables the inclusion of information with indeterminate timing.

Another distinctive inference capability facilitated by ETHOS is DRG class estimation. As illustrated in Fig. 5c, the token denoting the DRG class is consistently positioned following the discharge token and a Q token specifying the length of hospital stay. With 771 unique tokens available for this purpose, we infer the actual class by generating a probability array in the final network layer of the transformer for the DRG token. This array is then utilized to predict the classification's top-1 and top-2 accuracy metrics.

Statistical analysis

The performance of classification algorithms of binary tasks was assessed using Receiver Operating Curve Analysis (ROC). The ROC curves were fitted to experimental points using Gaussian models with unequal variances for binary hypotheses (code provided). Values of Areas Under Curves (AUCs) and 95% confidence intervals (CI) were calculated using bootstrapping (code provided). For multiclass classification (DRG task), we used top-1 and top-2 accuracy. We used mean absolute error (MEA) for the regression tasks to indicate prediction fidelity with 95% confidence intervals estimated using bootstrapping. Python numpy and scikit-learn were used.

Comparison of Ethos to existing methods

Employing the data segmentation as detailed in Supplementary Table 1, we evaluated traditional algorithms for predicting 30 day hospital readmission rates and juxtaposed these outcomes with those obtained via ETHOS. The features used in Supplementary Fig. 6 were culled from data accrued during the patient's hospitalization, adhering to the feature derivation methodology outlined by Tang et al.[23]. Attempting to apply the algorithm devised by the authors to our dataset presented challenges, notably due to the Graph Neural Network (GNN) implementation by Tang et al.[23], which necessitates the computation of a similarity score for each pair of admissions. Given the

可解释性。这种方法确保单个代币对网络结果的可解释性做出重大贡献。有关我们分析中应用于 MIMIC 数据的标记化过程的更多信息，以及患者健康时间线 (PHT) 的示例，请读者参阅补充表 3，其中我们提供了本工作中使用的真实 PHT 和注释。补充表 2 总结了 MIMIC 数据集的所有标记化组件。

精神培训

我们采用受 Transformer[3] 解码器架构启发的模型，将自然语言处理 (NLP) 中的标记化文本与我们的 PHT 标记化方法进行比较。我们的模型开发基于 Andrej Kapathy 的 GPT-2 实现 (github.com/karpathy/nanoGPT)。设计选择与原始 Transformer 论文略有不同，因为它不使用固定的正弦位置编码，而是利用可学习的位置嵌入，这些位置嵌入在令牌转换为相应嵌入的阶段添加到令牌嵌入中。ETHOS 模型的训练首先从现有患者记录中合成数据集。每个患者的 PHT 都以“时间线结束”标记结束，然后将它们连接起来，为训练创建一个长标记序列。与生成式 LLM 类似，ETHOS 被训练为根据先前令牌的上下文来预测单个令牌。考虑到大数据规模和模型复杂性，此阶段是资源密集型的，类似于 LLM[3,35] 中使用的 NLP 转换器的训练方法。我们估计 ETHOS 面临的网络训练任务的大小与 GPT-2[8] 类似，因此我们使用该网络中使用的变压器的大小作为起点（超参数搜索和选择的详细信息可以在补充图 2 中看到）。我们对网络的大小进行启发式调整以优化损失函数的值。 Brown 等人提供了有关 Transformer 训练方法的更多详细信息。[8] 中提供了关于我们的实现的补充材料和完整代码，发布于 https://github.com/ipolharvard/ethos-paper。

精神推论

在推理过程中，ETHOS 的功能类似于文档完成工具，其中单词序列而不是健康相关事件被排序到 PHT 中。该程序从患者 PHT 中记录的病史开始。最后 2048 个令牌（或整个 PHT（如果包含少于 2048 个令牌））用于在当前 ETHOS 实现中启动推理。然后，ETHOS 通过以下步骤一次生成一个令牌：(1) 为所有潜在令牌生成一组概率，(2) 根据这些概率随机选择一个新令牌，(3) 将新令牌附加到序列中，同时删除最旧的令牌以将上下文大小维持在 2048 个令牌，(4) 转到 1。此生成序列继续进行，直到遇到预定义的停止条件，其中可能包括显示患者的令牌的出现死亡或时间间隔标记的总和超过特定阈值。可以建立额外的停止标准。该方法的随机性允许创建多个未来 PHT (fPHT)。多个 fPHT 用于评估不确定性，因为每个 fPHT 都代表对未来的替代预测。

使用 ETHOS 评估临床结果和任务

选择这些实验是为了将结果与其他人的工作进行比较，以估计 MIMIC 数据的住院患者死亡率和再入院率。 MIMIC 中的患者被随机分为训练组和测试组，比例为 90%/10%（补充表 1）。

除非出院日期未知，否则在测试集中的所有住院患者入院时评估住院死亡率的可能性。这是通过生成过程执行的，该生成过程以准入令牌开始，以生成放电或死亡令牌结束，重复此循环 20 次。 “N”代表首先生成死亡令牌的次数，除以 20 来估计住院患者死亡的机会。同样，计算了 ICU 死亡的可能性

对于 MIMIC 数据集，还进行了一项额外的实验，其中预测是在 ICU 入院后 24 小时开始，而不是在 ICU 入院时开始。在同一模拟中，通过聚合模拟时间线中生成的时间间隔令牌（直到放电令牌出现）来估计 ICU 中的 LOS。模拟期间患者在 ICU 死亡的情况被排除在 LOS 计算之外。我们选择 20 次重复，产生 21 个独特的概率估计器，这足以构建稳健的接收器操作特性 (ROC) 曲线，产生出色的高斯拟合（图 3）。然而，也可以采用替代的重复计数。

为了计算 30 天住院患者再次入院的概率，fPHT 的生成从住院患者出院令牌开始，并在出现新的入院或死亡令牌或生成的累积时间令牌超过 30 天时停止。模拟重复了 20 次。然后得出 30 天再入院的概率为 M/20，其中“M”是由于 20 次重复中患者新入院标记而发生的终止计数。

在我们的方法中，任务是通过模拟未来患者健康时间表来完成的。然而，ETHOS 提供了额外的方法来获取见解，我们在这里举例说明其中的两种方法。例如，在每个 ICU 入院后构建 PHT 时，会创建一个序列，从标识 ICU 类型的标记开始，然后是 SOFA 分数标记，然后是表示第一天实际 SOFA 分数的 Q 标记。我们使用 ETHOS 生成的 SOFA Q 节点概率和标记化过程中分配的每个分位数的平均 SOFA 分数来预测 SOFA 分数（图 5a）。

数据集中没有指定 1 天 SOFA 分数评估的确切时间，入院后立即插入 SOFA 分数会导致潜在的因果关系问题，因为它依赖于随后获取的数据。在模型的训练阶段，ETHOS 允许这种明显的因果关系违规。然而，此类 1 天 SOFA 评分的真实值在入住 ICU 时无法获得，因此不会用于在推理过程中模拟未来时间线，以防止推理过程中违反因果关系。相反，正如我们的研究所示，这些分数是根据先前信息预测的。 ETHOS 的这一功能可以包含不确定时间的信息。

ETHOS 提供的另一个独特的推理能力是 DRG 类别估计。如图 5c 所示，表示 DRG 类别的标记始终位于出院标记和指定住院时间的 Q 标记之后。有 771 个可用于此目的的唯一令牌，我们通过在 DRG 令牌的转换器的最终网络层中生成概率数组来推断实际类别。然后使用该数组来预测分类的 top-1 和 top-2 准确度指标。

统计分析

使用接收器操作曲线分析（ROC）评估二元任务分类算法的性能。使用二元假设方差不等的高斯模型将 ROC 曲线拟合到实验点（提供代码）。曲线下面积 (AUC) 和 95% 置信区间 (CI) 的值是使用 bootstrapping 计算的（提供了代码）。对于多类分类（DRG 任务），我们使用 top-1 和 top-2 精度。我们在回归任务中使用平均绝对误差 (MEA) 来指示使用自举估计的 95% 置信区间的预测保真度。使用了 Python numpy 和 scikit-learn。

精神与现有方法的比较

我们采用补充表 1 中详述的数据分段，评估了预测 30 天医院再入院率的传统算法，并将这些结果与通过 ETHOS 获得的结果进行了比较。补充图 6 中使用的特征是从患者住院期间积累的数据中挑选出来的，遵循 Tang 等人概述的特征推导方法。[23]。尝试将作者设计的算法应用于我们的数据集提出了挑战，特别是由于 Tang 等人的图神经网络 (GNN) 实现。[23]，这需要计算每对录取的相似性分数。鉴于

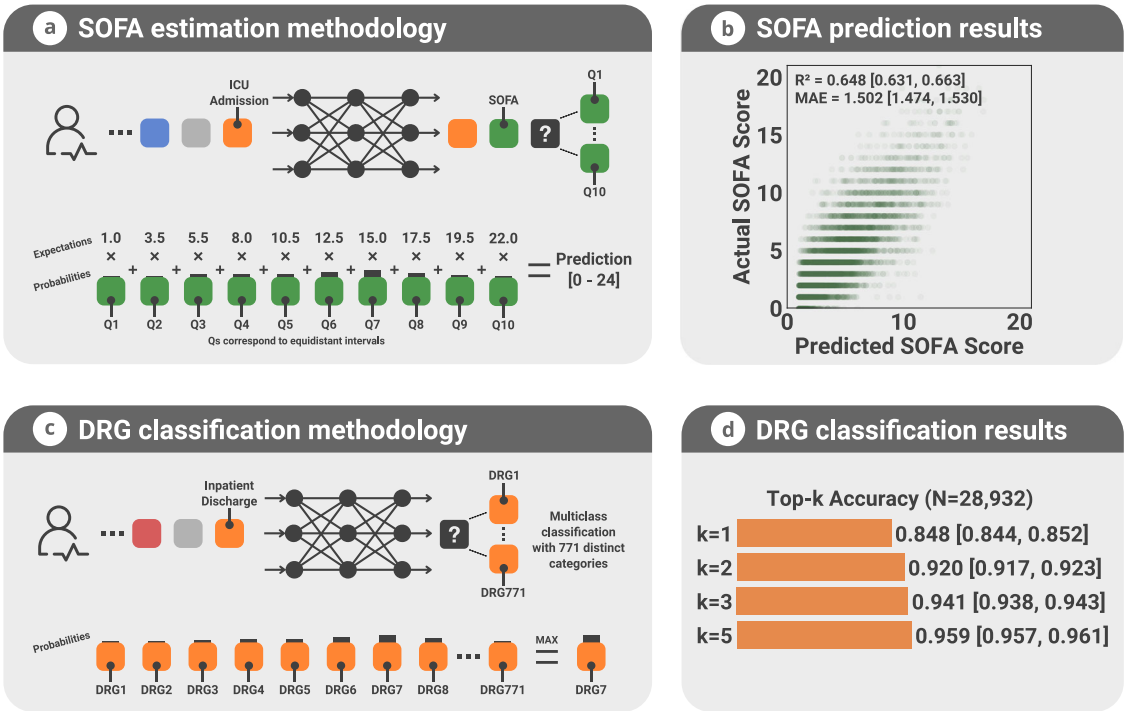

Fig. 5 | ETHOS model performance on SOFA estimation and DRG classification. **a** Estimation of the first-day Sequential Organ Failure Assessment (SOFA) score at ICU admission by ETHOS, which generates a sequence of three tokens: the admission type (orange token), a SOFA token (indicating the SOFA score estimation will follow), and a quantile token (q-token indicated by question mark) predicting probabilities of the SOFA score's quantile, as detailed at the bottom of the (**a**). The fixed position of the SOFA token ensures its consistent prediction immediately after ICU admission. The SOFA score is derived using quantile probabilities generated by ETHOS and average value of SOFA for ten quantiles (values of 1.0, 3.5 …). Since SOFA value 24 was not present in the dataset we predict values 0–23. **b** Correlation plot between actual and predicted SOFA scores. **c** For Diagnostic Related Groups (DRG) classification. The model is trained to insert a DRG token after tokens typically used at discharge time, utilizing a placeholder "DRG_UNKNOWN" for if DRG is unknown in the training set. Predicted probabilities are used to compute the top-{1,2,3,5} DRG classifications. **d** Visualization of DRG classification accuracy, showcasing the model's predictive performance.

significantly larger volume of admissions in our dataset—approximately 400,000, in stark contrast to the 14,500 reported by Tang et al.—this task proved impractical on a compute node with 2TB of RAM, defying all efforts to achieve it within a reasonable timeframe. Consequently, we limited our application to the data preprocessing and feature extraction segments of Tang et al.'s methodology (Tang et al. 2023)[23]. The adapted and modified code from Tang et al.'s repository, which we cloned for feature extraction, is accessible at github.com/ipolharvard/readmit-stgnn. For models unsupportive of temporal sequence analysis, such as Logistic Regression and XGBoost, we modified the approach to handle time-varying features by consolidating them over time. This entailed distilling the minimum, first quartile, median, third quartile, and maximum values of dynamically changing features. Furthermore, we integrated the day of admission as a unique feature to retain an element of temporal dimension within the dataset. In Supplementary Fig. 6 ETHOS was compared to one of the leading proprietary LLM models - GPT-4o in two temperature variants: 0.3 and 0.5. We constructed a comprehensive prompt that directs the model to analyze a timeline of 2048 tokens and calculate the probability of patient readmission for 2000 cases from the test set. This prompt is structured into four distinct parts: task instructions, a basic patient description corresponding to ETHOS's static information, PHT and a detailed description of subgroups of tokens and their identification. The complete codebase for this experiment including the prompt design is accessible at https://github.com/ipolharvard/ethos-paper/blob/master/notebooks/llm_readmission_task.ipynb. ETHOS significantly outperforms both variants of GPT-4o for the same subset of testing samples.

Data availability

The MIMIC-IV dataset is publicly available at https://physionet.org/content/mimiciv/2.2/.

Code availability

The code, ETHOS model weights used for all inferences, results of inferences, scripts to generate numerical results for all aspects of this study for the MIMIC-IV dataset are made publicly available at github.com/ipolharvard/ethos-paper. In our experiments, we used Python 3.10, and the following open-source libraries: torch = 2.3.0, joblib = 1.4.2, tqdm = 4.66.4, colorlog = 6.8.2, h5py = 3.11.0, pandas = 2.2.2, numpy = 1.26.4, pyarrow = 16.1.0, click = 8.1.7.

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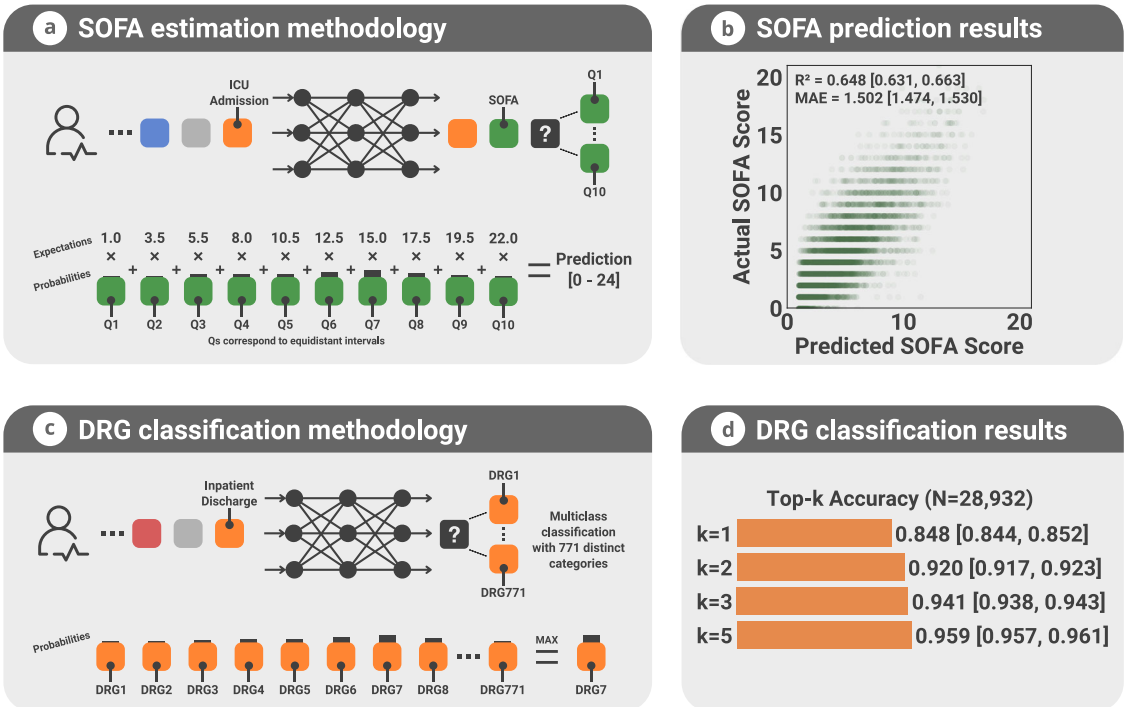

图 5 | ETHOS 模型在 SOFA 估计和 DRG 分类上的性能。 a ETHOS 对 ICU 入院第一天序贯器官衰竭评估 (SOFA) 评分进行估计，生成三个标记的序列：入院类型（橙色标记）、SOFA 标记（表示随后将进行 SOFA 评分估计）和分位数标记（由问号表示的 q 标记），用于预测 SOFA 评分分位数的概率，如 (a) 底部详述。 SOFA 代币的固定位置确保其在进入 ICU 后立即进行一致的预测。 SOFA 分数是使用以下生成的分位数概率得出的 ETHOS 和 SOFA 十个分位数的平均值（值 1.0、3.5 ⋯）。由于数据集中不存在 SOFA 值 24，我们预测值 0-23。 b 实际 SOFA 分数与预测 SOFA 分数之间的相关图。 c 用于诊断相关组 (DRG) 分类。该模型经过训练，可在放电时通常使用的标记后插入 DRG 标记，如果训练集中的 DRG 未知，则使用占位符“DRG_UNKNOWN”。预测概率用于计算 top-{1,2,3,5} DRG 分类。 d DRG 分类准确性的可视化，展示模型的预测性能。

我们的数据集中的录取量要大得多——大约 400,000 个，与 Tang 等人报告的 14,500 个形成鲜明对比——事实证明，这项任务在具有 2TB RAM 的计算节点上是不切实际的，无法在合理的时间范围内实现这一目标。因此，我们将应用限制在 Tang 等人的方法的数据预处理和特征提取部分（Tang 等人，2023）[23]。我们从 Tang 等人的存储库中克隆了经过改编和修改的代码以进行特征提取，可从 github.com/ipolharvard/readmit-stgnn 访问。对于不支持时间序列分析的模型，例如 Logistic 回归和 XGBoost，我们修改了处理随时间变化的特征的方法，方法是随着时间的推移对其进行整合。这需要提取动态变化特征的最小值、第一四分位数、中位数、第三四分位数和最大值。此外，我们将入院日期作为一个独特的特征进行整合，以保留数据集中的时间维度元素。在补充图 6 中，ETHOS 与领先的专有 LLM 模型之一 - GPT-4o 在两种温度变化下进行了比较：0.3 和 0.5。我们构建了一个全面的提示，指导模型分析 2048 个标记的时间线，并计算测试集中 2000 个病例的患者重新入院的概率。该提示由四个不同的部分组成：任务说明、与 ETHOS 静态信息相对应的基本患者描述、PHT 以及标记子组及其标识的详细描述。该实验的完整代码库（包括提示设计）可从 https://github.com/ipolharvard/ethos-paper/blob/master/notebooks/llm_readmission_task.ipynb 访问。对于相同的测试样本子集，ETHOS 的性能明显优于 GPT-4o 的两个变体。

数据可用性

MIMIC-IV 数据集可在 https://physionet.org/content/mimiciv/2.2/ 上公开获取。

代码可用性

用于所有推理的代码、ETHOS 模型权重、推理结果、为 MIMIC-IV 数据集本研究的各个方面生成数值结果的脚本均在 github.com/ipolharvard/ethos-paper 上公开提供。在我们的实验中，我们使用Python 3.10和以下开源库：torch = 2.3.0、joblib = 1.4.2、tqdm = 4.66.4、colorlog = 6.8.2、h5py = 3.11.0、pandas = 2.2.2、numpy = 1.26.4，pyarrow = 16.1.0，点击= 8.1.7。

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A.S. and P.R. conceptualized the work. A.S., P.R., Y.J. and A.E.S. designed the study. P.R. and A.S. performed the coding and the experiments. Y.J. and A.S. conducted the literature search. D.B., A.E.S., Q.L. and J.W. provided advisory support for the project. P.R. and A.S. prepared the initial draft of the manuscript, with all authors actively participating in the refinement and finalization of the manuscript through comprehensive review and contributions. A.S. supervised the project.

Competing interests

Y.J. is currently also affiliated with Verily life science, SSF, CA. The other authors declare no competing interests.

Additional information

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